ResiDual for Audio: Spectral Reweighting of Residual Streams in CLAP

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Abstract

Contrastive audio-language models such as CLAP achieve remarkable zero-shot classification performance, yet their internal representations remain largely unexplored. I investigate the structure of attention head outputs in HTS-AT, revealing a progressive emergence of semantically specialized representations across the network hierarchy, tightly linked to the intrinsic dimensionality of per-head activations. These insights motivate the ResiDual* method, a parameter-efficient method that reweights the principal components of each attention head without modifying any pretrained weight. Evaluated under 5-fold cross-validation on four benchmarks, ResiDual* substantially improves over the frozen CLAP baseline.

1. Introduction

Contemporary audio–text encoders excel at capturing multimodal correspondences (Manco et al., 2022; Elizalde et al., 2022), yet the spectral structure expressed through attention heads remains under-analysed. Recent findings indicate that attention heads give rise to highly low-dimensional subspaces (Wang et al., 2025; Basile et al., 2024), suggesting latent geometric constraints that audio-domain Transformers do not explicitly exploit. This motivates an investigation into whether analogous structures emerge in HTS-AT (Chen et al., 2022) — a hierarchical Swin-Transformer variant that serves as the audio encoder in CLAP, mapping raw waveforms to a shared audio-text embedding space — and whether spectral selectivity can serve as an effective adaptation strategy for improving zero-shot classification.

In response, I introduce RESIDUAL*, a spectral reweighting framework extending the residual-subspace methodology of ResiDual to the audio domain. The method (i) analyses attention-head dimensionality and semantic specialization across all 184 HTS-AT heads, revealing how representational complexity evolves across the network hierarchy and which stages are most amenable to spectral intervention; and (ii) yields measurable zero-shot classification improvements by learning, on a small labelled partition, a per-component rescaling of each attention head's principal directions — amplifying task-relevant spectral components and suppressing noise — without modifying any pretrained parameter. Full architectural details of CLAP and HTS-AT are provided in Appendix A; the complete head analysis pipeline and results in Appendix B; the ResiDual* implementation in Appendix C; and the full evaluation protocol and per-fold results in Appendix D. The appendices document every step of the research process — including design choices, intermediate experiments, and the reasoning behind the final method.

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2. Related Works

Spectral Decomposition in Transformers. The ResiDual framework formalizes residual streams through eigenspace decomposition, showing that attention heads operate within constrained, low-dimensional subspaces capturing specific semantic roles. This aligns with Voita et al. (Voita et al., 2019), who showed that only a subset of heads is critical for performance while others can be pruned with minimal impact.

Audio–Text Models and CLAP. CLAP has emerged as a leading architecture for joint audio-text embeddings, building on CLIP-like methods (Radford et al., 2021) and attention-based audio models (Gong et al., 2021). Despite these advances, internal representation geometry within residual pathways remains underexplored: prior studies examined attention distributions (Yang et al., 2020; Wu et al., 2020; Won et al., 2019) or audio embeddings (Zhang et al., 2025a), but not the spectral properties of residual streams. To the best of my knowledge, this is the first systematic spectral analysis and reweighting strategy applied to CLAP.

Spectral Reweighting and Representation Geometry. Pretrained models are known to produce anisotropic representations, degrading cosine-similarity-based comparisons (Ethayarajh, 2019). Post-hoc corrections such as All-but-the-Top (Mu et al., 2017) and whitening (Su et al., 2021) address this at the output embedding level. ResiDual showed that reweighting principal components of attention head outputs — rather than output embeddings — yields more targeted spectral alignment. I extend this paradigm to the audio domain, applying it to the HTS-AT within CLAP.

3. Method

Baseline. The baseline is CLAP used off-the-shelf in a fully frozen, zero-shot regime. Given an audio clip x, the ℓ_2-normalised audio embedding $\hat{\mathbf{a}}(x) \in \mathbb{R}^{d_{\text{proj}}}$ is compared against the ℓ_2-normalised text embeddings $\{\mathbf{t}_c\}$ of class prompts `"this is the sound of {c}"`, and the predicted class is $\hat{y} = \arg\max_c \hat{\mathbf{a}}(x)^\top \mathbf{t}_c$[1]. No adaptation, fine-tuning, or labelled data is used.

Contribution: ResiDual*. ResiDual* adapts the frozen CLAP model by learning a per-component spectral reweighting of each attention head in HTS-AT, without modifying any pretrained parameter.

Head Analysis. To characterise the internal geometry of HTS-AT, I extract the raw pre-projection head outputs $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$ for all 184 heads and spatially pool them into per-sample vectors:

$$\mathbf{r}_{\ell,b,h} = \frac{1}{N_w^\ell M} \sum_{i,j} \mathbf{H}_{\ell,b,h}[i,j,:] \in \mathbb{R}^{d_h}. \quad (1)$$

I estimate intrinsic dimensionality via both linear (PCA, Effective Rank) and nonlinear (TwoNN, MLE) estimators, and measure semantic specialization through a cross-modal grounding framework that projects each head's principal axis into the CLAP embedding space and decomposes it via Orthogonal Matching Pursuit (Pati et al., 1993) over a textual dictionary (see Appendix B).

Fitting, Optimisation, and Reweighting. For each target stage $\ell \in \mathcal{L}$, block b, and head h, I estimate a PCA basis $\mathbf{\Phi}_{\ell,b,h}$ and mean $\boldsymbol{\mu}_{\ell,b,h}$ from audio clips via forward hooks, and initialise a learnable weight vector $\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$. The vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ are then optimised via cross-entropy minimisation on a small labelled partition. At inference time, each head output is transformed as:

$$\mathrm{RD}^*(\mathbf{H}_{\ell,b,h}) = \left(\mathbf{H}_{\ell,b,h} - \boldsymbol{\mu}_{\ell,b,h}\right) \mathbf{\Phi}_{\ell,b,h}\ \mathrm{diag}(\boldsymbol{\lambda}_{\ell,b,h})\ \mathbf{\Phi}_{\ell,b,h}^\top.$$

Full implementation details and hyperparameters are reported in Appendix C and Table 3.

[1]Since both vectors have unit norm, the dot product reduces to cosine similarity.

4. Results

Head Analysis. The dimensionality and semantic coherence analysis across all 184 heads (see Appendix B) reveals a clear structural transition: early-stage heads operate in near-unidimensional regimes (EVR_1 between 0.67 and 0.92 in Stage 0), while Stages 2 and 3 develop high-dimensional, semantically coherent manifolds, with coherence Z-scores peaking at $Z = 1.57$ for heads such as `L3_B0_H7` (*human_body*) and `L2_B5_H15` (*nature*) (Figure 5, Table 4). This motivates restricting the reweighting target to $\mathcal{L} = \{2, 3\}$.

Zero-Shot Classification. Figure 1 reports mean 5-fold zero-shot accuracy across ESC-50 (Piczak), IRMAS (Juan J. Bosch & Herrera), TinySOL (Emanuele et al., 2020), and VocalSound (Gong et al., 2022) (full per-fold results in Table 5 and Table 6, Appendix D; hyperparameters in Table 3). ResiDual* consistently and substantially outperforms the frozen CLAP baseline on all benchmarks, with gains ranging from +2.4 pp on ESC-50 to +54.8 pp on TinySOL (see also Figure 9, Appendix D). The improvement is most pronounced on fine-grained timbral tasks (TinySOL, IRMAS), where the generic audio-language alignment of CLAP is insufficient, and the spectral reweighting successfully re-aligns internal representations toward task-relevant subspaces.

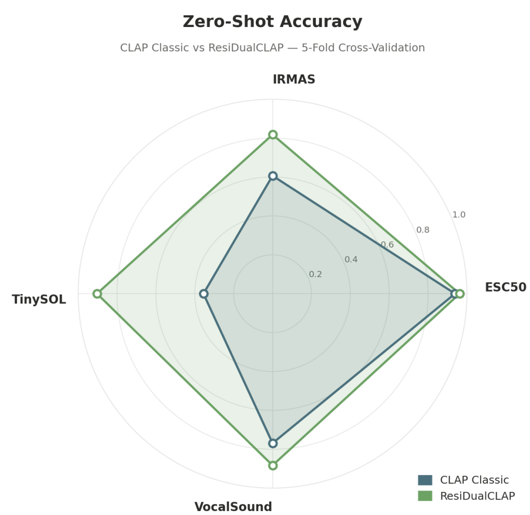

Figure 1. Mean 5-fold zero-shot accuracy across benchmarks. CLAP Classic (blue) vs. ResiDual* (green).

This pattern is consistent with the nature of the task: fine-grained datasets such as TinySOL and IRMAS are characterised by classes defined by highly distinctive timbral properties, localised in specific spectral regions. In such settings, a mechanism that selectively amplifies informative principal directions proves substantially more effective than the global, generic representations produced by the frozen CLAP encoder.

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A. CLAP and HTS-AT: Architecture, Notation, and Residual Decomposition

This appendix is a self-contained reference for the two components at the core of the system. We begin by describing the CLAP dual-encoder model and its configuration (§A.1). We then present the HTS-AT audio backbone in detail, covering input preprocessing, hierarchical stage structure, and the internal mechanics of each attention block (§A.2). We then derive the per-head residual decomposition that underlies my analysis (§A.3). Finally, I provide implementation details for the ResiDual spectral reweighting (Appendix C). All notation is introduced inline and collected in Table 3 for reference.

A.1. CLAP Overview

CLAP (Contrastive Language–Audio Pretraining) (Elizalde et al., 2022) is a dual-encoder model that aligns audio and text representations in a shared embedding space via contrastive learning. The main audio encoder is HTS-AT (Chen et al., 2022) and the main text encoder is GPT-2 (Radford et al., 2019) (embedding dimension 768), whose weights are frozen throughout training. Both encoders are independently projected to a joint space of dimension $d_{\text{proj}} = 1024$ via dedicated linear projection heads; similarity is measured by temperature-scaled cosine similarity with InfoNCE loss at temperature $\tau = 0.003$. Zero-shot audio classification is performed by comparing an audio embedding against the embeddings of textual class prompts. Given a set of class labels $\{c_1, \ldots, c_C\}$, each label is wrapped in a prompt template (e.g. *"the sound of* $\{$*class*$\}$*"*) and encoded by the text encoder to obtain ℓ_2-normalised embeddings $\mathbf{t}_c \in \mathbb{R}^{d_{\text{proj}}}$. For an audio clip x, the audio encoder followed by the projection head P produces the ℓ_2-normalised embedding $\hat{\mathbf{a}}(x) \in \mathbb{R}^{d_{\text{proj}}}$. The predicted class is then

$$\hat{y}(x) = \arg\max_{c \in \{c_1, \ldots, c_C\}} \hat{\mathbf{a}}(x)^\top \mathbf{t}_c, \tag{2}$$

i.e. the class whose text embedding is most similar to the audio embedding under cosine similarity (the temperature τ cancels under $\arg\max$ and is omitted). Zero-shot accuracy is then computed as the fraction of correctly classified clips over the evaluation set, without any fine-tuning of the model parameters.

The full configuration used in this work is reported in Table 1.

A.2. HTS-AT Architecture

HTS-AT (Chen et al., 2022) is a hierarchical Transformer architecture based on the Swin Transformer (Liu et al., 2021), adapted to operate on audio spectrograms. Its computation is organised into four hierarchical *stages*, each

Table 1. CLAP configuration parameters.

Parameter	Value
Text encoder	GPT-2
Text encoder embedding dim	768
Audio encoder	HTS-AT
Audio encoder output dim (D_3)	768
Joint projection dim (d_{proj})	1024
Contrastive temperature τ	0.003
Sampling rate	44 100 Hz
Audio duration	7 s
Mel bands	64
FFT window	1024
Hop size	320
$f_{\min}$ / $f_{\max}$	50 Hz / 8 000 Hz

composed of several *blocks*, as illustrated in Figure 2. I describe the pipeline from raw audio to the final embedding in order: input preprocessing, patch embedding, stage structure, and the attention mechanism inside each block.

Input preprocessing and patch embedding. Each audio clip is converted to a 64-band log-mel spectrogram ($f_{\min} = 50\,\text{Hz}$, $f_{\max} = 8{,}000\,\text{Hz}$, STFT window 1024, hop 320) and normalised per mel-band. The resulting time–frequency matrix is rearranged into a square image $\mathbf{x}_{\text{img}} \in \mathbb{R}^{1\times256\times256}$ by folding the time axis into four contiguous segments of 256 frames stacked vertically over the 64 mel-frequency bands ($4\times64 = 256$ rows, 256 columns). This image is then split into 4×4 non-overlapping patches via a strided `Conv2d` layer (kernel 4×4, stride 4), followed by `LayerNorm`, yielding the input token sequence

$$\mathbf{Z}^{(0,0)} = \mathbf{X}_{\text{in}} \in \mathbb{R}^{4096\times96}, \tag{3}$$

where $4096 = 64 \times 64$ tokens each have embedding dimension $D_0 = 96$.

Stage structure and PatchMerging. The four stages operate at progressively coarser spatial resolutions, as summarised in Table 2. We index the residual stream as $\mathbf{Z}^{(\ell,b)}$, where $\ell \in \{0, 1, 2, 3\}$ is the stage index and $b \in \{1, \ldots, B_\ell\}$ is the block index within that stage.

After Stages 0, 1, and 2, a *PatchMerging* layer concatenates each 2×2 neighbourhood of spatially adjacent tokens into a single vector of dimension $4D_\ell$, then projects it to $2D_\ell$ via a bias-free linear layer. This halves the spatial side length S_ℓ and doubles the embedding dimension:

$$\mathbb{R}^{S_\ell^2\times D_\ell} \xrightarrow{\text{PatchMerging}} \mathbb{R}^{(S_\ell/2)^2\times 2D_\ell}. \tag{4}$$

Stage 3 has no PatchMerging. The resulting spatial side lengths are $S_\ell \in \{64, 32, 16, 8\}$ and the corresponding embedding dimensions are $D_\ell \in \{96, 192, 384, 768\}$ (see Table 2).

Within-stage residual structure. Inside each stage ℓ, both the attention and MLP sub-layers write additively to the residual stream. Following the pre-norm formulation (Wang et al., 2019), the stream after block b is:

$$\mathbf{Z}^{(\ell,b)} = \mathbf{Z}^{(\ell,0)} + \sum_{b'=1}^{b} \mathbf{A}_{\ell,b'} + \sum_{b'=1}^{b} \mathbf{M}_{\ell,b'}, \quad \in \mathbb{R}^{S_\ell^2\times D_\ell}, \tag{5}$$

where $\mathbf{Z}^{(\ell,0)}$ is the stage input, $\mathbf{A}_{\ell,b}$ is the W-MSA output, $\mathbf{M}_{\ell,b}$ is the MLP output, and S_ℓ is the spatial side length of the token grid at stage ℓ.

This decomposition holds strictly *within* a stage, where D_ℓ is constant. Across stage boundaries, PatchMerging changes both spatial resolution and embedding dimension ($S_\ell \to S_\ell/2$, $D_\ell \to 2D_\ell$), breaking any global additive structure across the full network, unlike in isotropic vision transformers.

Per-head output of the attention layer. The W-MSA output $\mathbf{A}_{\ell,b}$ is produced by concatenating the outputs of H_ℓ independent attention heads and projecting through a shared output matrix $W^O_{\ell,b}$. At block (ℓ, b), the raw output of head h is:

$$\mathbf{H}_{\ell,b,h} = \text{Softmax}\left(\frac{\mathbf{Q}_h\,\mathbf{K}_h^\top}{\sqrt{d_h}} + \mathbf{B}_{\ell,b,h}\right)\mathbf{V}_h \;\in\; \mathbb{R}^{N_w^\ell\times M\times d_h}, \tag{6}$$

where $N_w^\ell = S_\ell^2/M$ is the number of attention windows, $M = 64$ tokens per window, and $\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M\times M}$ is the learned relative position bias for head h at block (ℓ, b) (shared across windows, independent per head and per block). All H_ℓ head outputs are concatenated and projected to give the W-MSA output (see §A.3 for the full decomposition):

$$\mathbf{A}_{\ell,b} = \left[\mathbf{H}_{\ell,b,1}\|\cdots\|\mathbf{H}_{\ell,b,H_\ell}\right] W^O_{\ell,b} + \mathbf{b}^O_{\ell,b} \;\in\; \mathbb{R}^{N_w^\ell\cdot M\times D_\ell}. \tag{7}$$

I analyse the raw head outputs $\mathbf{H}_{\ell,b,h}$ directly, *before* the output projection $W^O_{\ell,b}$. Each $\mathbf{H}_{\ell,b,h}$ lives in the per-head space $\mathbb{R}^{d_h}$, which is constant across all stages and blocks. This choice avoids entangling information across heads introduced by the shared projection, and enables uniform cross-stage comparisons of head geometry throughout the network. A complementary analysis of the projected contributions $\widehat{\mathbf{H}}_{\ell,b,h}$—which characterize how each head shapes the residual stream—was conducted in the early stages of the analysis but is not reported here, as no meaningful differences were observed between the pre- and post-projection representations. This is one of several differences with respect to the original ResiDual approach.

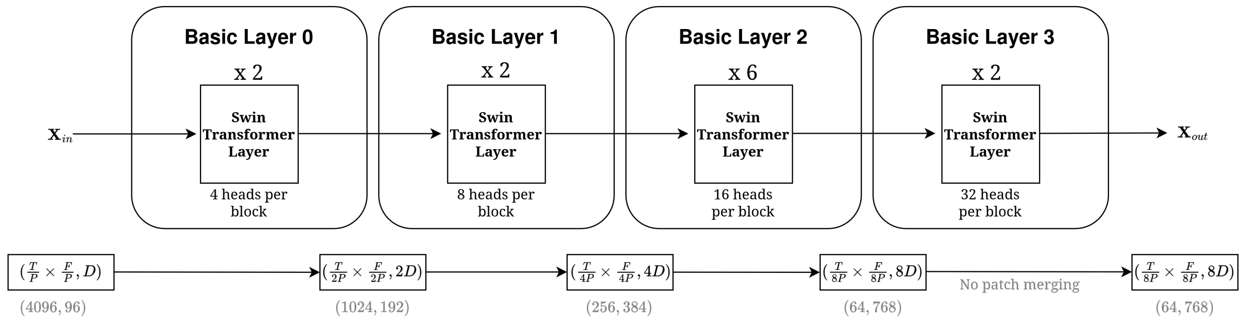


Figure 2. HTS-AT hierarchical architecture. The model consists of four stages of increasing embedding dimension: Stage 0 (2 blocks, 4 heads), Stage 1 (2 blocks, 8 heads), Stage 2 (6 blocks, 16 heads), and Stage 3 (2 blocks, 32 heads). PatchMerging is applied after Stages 0, 1, and 2, halving the spatial side length and doubling the feature dimension at each transition. Stage 3 has no PatchMerging. The input token sequence has spatial size $64 \times 64 = 4{,}096$ tokens with embedding dimension $D_0 = 96$; after three PatchMerging operations the final spatial size is $8 \times 8 = 64$ tokens with $D_3 = 768$.

Table 2. HTS-AT stage-level architectural parameters. The per-head dimension $d_h = D_\ell / H_\ell = 24$ is constant across all stages. S_ℓ is the spatial side length of the token grid at stage ℓ; $N_w^\ell = S_\ell^2 / M$ is the number of attention windows, with $M = w^2 = 64$ tokens per window ($w = 8$).

Stage ℓ	**Blocks** B_ℓ	**Heads** H_ℓ	**Dim** D_ℓ	**Spatial** S_ℓ^2	**Windows** N_w^ℓ
0	2	4	96	64×64	64
1	2	8	192	32×32	16
2	6	16	384	16×16	4
3	2	32	768	8×8	1
Total heads H_{tot}		$2{\cdot}4 + 2{\cdot}8 + 6{\cdot}16 + 2{\cdot}32 = 8 + 16 + 96 + 64 = 184$			

Block computation: W-MSA and MLP. Each block b at stage ℓ applies two sub-layers in sequence, both preceded by LayerNorm and connected via residual additions:

$$\mathbf{Z}^{(\ell,b)} \leftarrow \mathbf{Z}^{(\ell,b-1)} + \text{W-MSA}_{\ell,b}\Big(\text{LN}\Big(\mathbf{Z}^{(\ell,b-1)}\Big)\Big), \quad (8)$$

$$\mathbf{Z}^{(\ell,b)} \leftarrow \mathbf{Z}^{(\ell,b)} + \text{MLP}_{\ell,b}\Big(\text{LN}\Big(\mathbf{Z}^{(\ell,b)}\Big)\Big). \quad (9)$$

Even-indexed blocks use standard Window Multi-head Self-Attention (W-MSA); odd-indexed blocks use Shifted-Window MSA (SW-MSA), which shifts the partition by $(w/2, w/2)$ tokens to enable cross-window interactions. At Stage 3, where $S_3 = w = 8$ so the grid coincides with a single window, the shift degenerates to zero and all blocks use W-MSA. The MLP sub-layer is a two-layer feed-forward network with hidden dimension $4D_\ell$ and GELU activation.

WindowAttention in detail. The W-MSA module at block (ℓ, b) first partitions the S_ℓ^2 tokens into $N_w^\ell = S_\ell^2/M$ non-overlapping windows of $M = 64$ tokens each, then applies multi-head attention independently within each window.

In the SW-MSA variant, a cyclic shift is applied before window partitioning.

Queries, keys, and values are computed via a single fused projection $W_{\ell,b}^{QKV} \in \mathbb{R}^{D_\ell \times 3D_\ell}$:

$$\text{LN}(\mathbf{Z}^{(\ell,b-1)})\, W_{\ell,b}^{QKV} \xrightarrow{\text{split}(D_\ell)} \mathbf{Q}, \mathbf{K}, \mathbf{V} \;\in\; \mathbb{R}^{N_w^\ell \times M \times D_\ell}. \quad (10)$$

Each tensor is then reshaped to isolate the H_ℓ heads, yielding per-head slices $\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h \in \mathbb{R}^{N_w^\ell \times M \times d_h}$ with $d_h = D_\ell / H_\ell = 24$.

For each head $h \in \{1, \ldots, H_\ell\}$, attention is computed as:

$$\mathbf{H}_{\ell,b,h} = \text{Softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}} + \mathbf{B}_{\ell,b,h}\right) \mathbf{V}_h \;\in\; \mathbb{R}^{N_w^\ell \times M \times d_h}, \quad (11)$$

where $\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M \times M}$ is the learned relative position bias for head h of block (ℓ, b). This bias encodes the relative spatial offset between each pair of tokens *within* a window; it is shared across all N_w^ℓ windows (broadcast) and is independent for each head. Concretely, it is read from a parameter table of shape $((2w-1)^2, H_\ell) = (225, H_\ell)$ via a fixed index mapping, so each head h has its own column of learnable values.

All H_ℓ head outputs are then concatenated and passed through the block-specific output projection $W_{\ell,b}^{O} \in$

$\mathbb{R}^{D_\ell \times D_\ell}$ with bias $\mathbf{b}^O_{\ell,b} \in \mathbb{R}^{D_\ell}$, giving the W-MSA output

$$\mathbf{A}_{\ell,b} = \left[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\right] W^O_{\ell,b} + \mathbf{b}^O_{\ell,b} \;\in\; \mathbb{R}^{N^\ell_w \cdot M \times D_\ell}, \tag{12}$$

which is added to the residual stream in Eq. (8).

A.3. Per-Head Residual Decomposition

The additive residual structure of Eq. (8) allows us to decompose the W-MSA contribution $\mathbf{A}_{\ell,b}$ exactly into independent per-head terms. Denoting by $W^O_{\ell,b,h} \in \mathbb{R}^{d_h \times D_\ell}$ the row slice of $W^O_{\ell,b}$ corresponding to head h (rows $[(h-1)d_h,\ hd_h)$), we distribute the output projection over heads:

$$\begin{aligned}\mathbf{A}_{\ell,b} &= \left[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\right] W^O_{\ell,b} + \mathbf{b}^O_{\ell,b} \\ &= \sum_{h=1}^{H_\ell} \left(\mathbf{H}_{\ell,b,h}\, W^O_{\ell,b,h} + \frac{\mathbf{b}^O_{\ell,b}}{H_\ell} \right) \;=\; \sum_{h=1}^{H_\ell} \widehat{\mathbf{H}}_{\ell,b,h}, \end{aligned} \tag{13}$$

where we define the *per-head projected contribution*

$$\widehat{\mathbf{H}}_{\ell,b,h} \;=\; \mathbf{H}_{\ell,b,h}\, W^O_{\ell,b,h} + \frac{\mathbf{b}^O_{\ell,b}}{H_\ell} \;\in\; \mathbb{R}^{N^\ell_w \times M \times D_\ell}. \tag{14}$$

This decomposition is exact and follows from the linearity of matrix multiplication: because $W^O_{\ell,b}$ acts on the concatenation of all head outputs, each head h effectively multiplies only its own row slice $W^O_{\ell,b,h}$, and the output bias can be distributed equally among heads without any approximation.

The decomposition holds within a single stage, where D_ℓ is constant. Across stage boundaries, PatchMerging changes both spatial resolution and embedding dimension, breaking any global additive structure; cross-stage comparisons must therefore account for this.

B. Head Analysis

B.1. Spatial Aggregation and Dataset Representations

Each raw head output $\mathbf{H}_{\ell,b,h}$ is a three-dimensional tensor indexed by window, token position, and feature dimension. To extract the attention head representations, I used a PyTorch hook attached to all the `WindowAttention` objects in CLAP. To obtain a single fixed-size vector per audio sample, I mean-pool[2] over all windows and token positions:

$$\mathbf{r}_{\ell,b,h} = \frac{1}{N^\ell_w\, M} \sum_{i=1}^{N^\ell_w} \sum_{j=1}^{M} \mathbf{H}_{\ell,b,h}[i,j,:] \;\in\; \mathbb{R}^{d_h}. \tag{15}$$

[2] Mean pooling was adopted primarily for computational tractability, as it reduces each head output to a fixed-size vector regardless of the number of windows N^ℓ_w.

This yields one vector $\mathbf{r}_{\ell,b,h}$ per audio sample, per head h, at every block (ℓ, b) of the network. Stacking these vectors across the n samples in the dataset gives the matrix

$$\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}, \tag{16}$$

which is the primary object of our analysis. The total number of such matrices across the model is $H_{\text{tot}} = 184$ (one per head per block; see Table 2).

Since $d_h = 24$ is constant across all stages, the ambient dimension of $\mathbf{R}_{\ell,b,h}$ is uniform throughout the network, enabling direct cross-stage comparisons of all dimensionality metrics without correction for a varying ceiling.

B.2. Dataset Representations

The dimensionality analysis is conducted on the ESC-50 dataset (Piczak), a benchmark collection of 2,000 environmental audio recordings organised into 50 semantic classes (e.g. *dog barking*, *rain*, *chainsaw*), each five seconds long and sampled at 44.1 kHz. ESC-50 was chosen for its compactness and semantic richness: the moderate number of samples ensures tractable computation of all dimensionality estimators, while the diversity of acoustic categories provides a sufficiently varied manifold to probe the representational capacity of individual heads.

For each recording, I extract the set of head representations $\{\mathbf{r}^{(i)}_{\ell,b,h}\}$ as described in §B.1, yielding matrices $\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$ with $n = 2{,}000$ and $d_h = 24$. All estimators detailed in §B.3 are applied independently to each of the $H_{\text{tot}} = 184$ head matrices.

B.3. Intrinsic Dimensionality Analysis

To characterize the effective complexity of the projected head representations $\{\mathbf{r}^{(i)}_{\ell,b,h}\}_{i=1}^{n} \subset \mathbb{R}^{1024}$, I employ a battery of linear and nonlinear dimensionality estimators.

B.3.1. Linear Estimators

PCA-based dimensionality. For each head, let $\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$ stack all aggregated representations. I compute the covariance matrix $\mathbf{C}_{\ell,b,h} = \frac{1}{n-1}\mathbf{R}^\top \mathbf{R}$ and obtain ordered eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots$. Linear intrinsic dimensionality is:

$$d_{\text{PCA}\alpha} = \arg\min_k \left\{ \frac{\sum_{i=1}^k \lambda_i}{\sum_i \lambda_i} \geq \alpha \right\}, \tag{17}$$

evaluated at $\alpha \in \{0.90, 0.95, 0.99\}$. I also report the *Explained Variance Ratio* of the first principal component, $\text{EVR}_1 = \lambda_1 / \sum_i \lambda_i$.

Participation Ratio.

$$\mathrm{PR} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}. \tag{18}$$

High PR signals uniform variance distribution; low PR signals dominance of few directions.

Effective Rank.

$$\mathrm{EffRank} = \exp\left(-\sum_i p_i \log p_i\right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}, \tag{19}$$

which can be interpreted as the exponential of the spectral entropy. Unlike PCA thresholds, it provides a smooth, threshold-free notion of dimensionality, capturing how uniformly variance is distributed across components.

B.3.2. Nonlinear Estimators

TwoNN (Facco et al., 2017).

$$d_{\mathrm{TwoNN}} = \left(\frac{1}{n}\sum_{i=1}^{n} \log \frac{r_2^{(i)}}{r_1^{(i)}}\right)^{-1}, \tag{20}$$

where $r_1^{(i)}, r_2^{(i)}$ are Euclidean distances to the first and second nearest neighbours of $\mathbf{r}_{\ell,b,h}^{(i)}$.

MLE (Levina & Bickel, 2004).

$$\hat{d}_{\mathrm{MLE}}(\mathbf{r}) = \left(\frac{1}{k-1}\sum_{j=1}^{k-1} \log \frac{r_k(\mathbf{r})}{r_j(\mathbf{r})}\right)^{-1}, \tag{21}$$

averaged over all samples, with $k = 20$. This estimator is based on a local maximum likelihood principle, assuming uniform sampling in small neighbourhoods, and is sensitive to fine-scale manifold structure beyond global linear approximations.

Linear-Nonlinear (L/N) Ratio

$$\mathrm{Ratio}_B = \frac{\bar{d}_{\mathrm{PCA}_{99}}}{\bar{d}_{\mathrm{TwoNN}}}. \tag{22}$$

Values near 1 indicate near-linear manifolds; higher values signal nonlinear curvature beyond what PCA captures, and serve as a diagnostic for selecting layer targets for spectral reweighting.

B.4. Results: Dimensionality Evolution Across the Hierarchy

The dimensionality analysis reveals a structured transition in how acoustic information is processed across the blocks of HTS-AT. Aggregate results per block are reported in Figure 3, while per-head detail is illustrated in Figure 4.

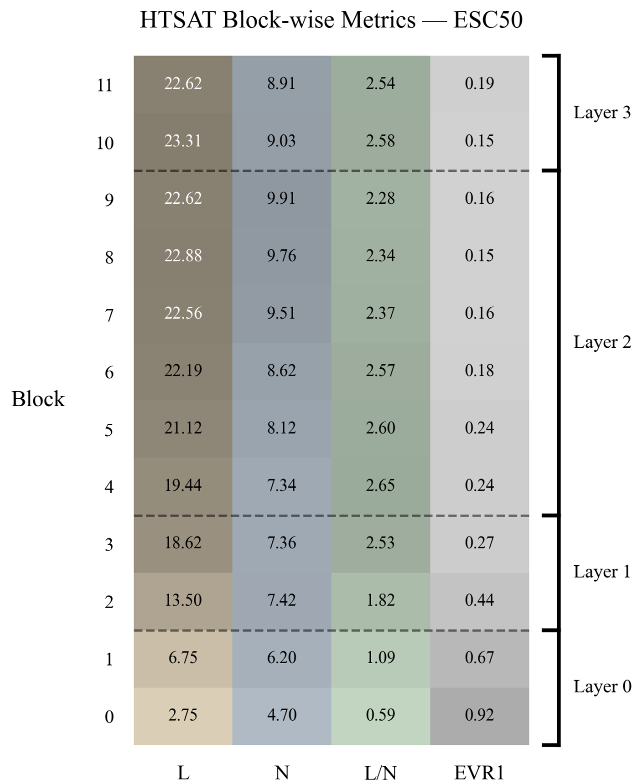


Figure 3. **Block-wise metrics for the ESC-50 dataset.** The heatmap shows the per-block average (blocks 0–11) of linear dimensionality (L with $\alpha = 0.99$), nonlinear dimensionality (N), the L/N ratio, and the dominance of the first principal component (EVR_1). Dashed lines separate the four stages (Layers 0–3) of the architecture.

B.4.1. Global Trends and Stage-wise Progression

Inspecting Figure 3, representational complexity clearly increases as we ascend the model hierarchy.

- **Layer 0 (Expansion Phase):** In the first blocks, intrinsic dimensionality is minimal ($L = 2.75$ in block 0). The very high EVR_1 value indicates that almost all variance is concentrated along a single direction. This suggests that the early stages act as low-complexity feature extractors, focusing on dominant signals in the time–frequency domain.
- **Layers 1 & 2 (Complexity Growth):** A rapid growth of both L and N is observed. The L/N ratio reaches its peak in Layer 2 (2.65), indicating that manifold curvature is maximal in this regime. Representations not only span more dimensions, but do so in a significantly nonlinear fashion relative to PCA bases.
- **Layer 3 (Saturation and Refinement):** In the final

blocks (10–11), dimensionality tends to stabilise or undergo a general slight contraction (exactly as happens in the ResiDual paper (Basile et al., 2024) in Figure 2.). EVR_1 reaches its historical minimum (0.15), indicating an extremely uniform distribution of variance across all available components ($d_h = 24$).

B.4.2. Head-specific Analysis and Manifold Structure

The per-head analysis (Figure 4) reveals the functional heterogeneity within the same block.

The plots highlight several key phenomena:

- **Saturation of Linear Capacity:** Panel (A) shows that many heads in Layers 2 and 3 reach the maximum dimensionality permitted by the $\alpha = 99\%$ threshold, saturating near the ambient dimension limit ($d_h = 24$).
- **Effective Rank vs. PR:** Both the Effective Rank (E) and the Participation Ratio (D) confirm a transition from a low-rank regime (Layer 0) to a higher-rank regime in the deeper layers, consistent with the increasing nonlinearity observed via TwoNN. The greater dispersion across heads in Layer 2 indicates geometric heterogeneity within the same stage: some heads operate in a lower-dimensional subspace while others span a higher-dimensional one. Notably, the heads exhibiting the highest semantic coherence (Table 4) tend to have lower PC1 variance, suggesting that concentrated spectral structure may be associated with more interpretable specialisation, though a systematic correlation between rank and semantic coherence remains to be established.
- **PC1 Dominance Decay:** Panel (F) illustrates the "deconstruction" of spectral dominance. Whereas the first heads (Layer 0) have a PC1 explaining nearly 100% of the variance, in the final heads this value drops below 20%, signalling distributed and potentially more robust representations.

Reflections on Representational Complexity. This evolution suggests that HTS-AT performs a progressive *unfolding* of the audio data: from nearly one-dimensional initial projections (likely tied to raw energetic or spectral features) toward complex, multidimensional manifolds in the intermediate layers, where semantic discrimination of ESC-50 classes takes place. The high L/N ratio in Layers 2 and 3 suggests that these blocks may be promising targets for spectral regularisation interventions, as they exhibit the greatest discrepancy between linear and nonlinear dimensionality estimates — a potential indicator of exploitable manifold curvature.

B.5. Head Specialization Analysis

To interpret the semantic content encoded by individual heads, I use a *cross-modal grounding* framework based on a pre-trained CLAP model. The goal is to map the principal variance axis of each head into a space of descriptive textual concepts.

B.5.1. Semantic Mapping via Principal Component Projection

Let $\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$ be the matrix of aggregated representations for a given head. To identify the primary acoustic feature captured by the head, I compute the first principal component $\mathbf{v}_1 \in \mathbb{R}^{d_h}$. For each sample i, the score on the first component is defined as:

$$s_i = \mathbf{r}_{\ell,b,h}^{(i)} \cdot \mathbf{v}_1. \tag{23}$$

I then define the **Semantic Direction** $\sigma_{\ell,b,h}$ in the CLAP latent space as the linear combination of CLAP audio embeddings $\{\mathbf{a}_i\}_{i=1}^n \subset \mathbb{R}^{D_{\mathrm{CLAP}}}$, weighted by the normalised scores:

$$\sigma_{\ell,b,h} = \sum_{i=1}^{n} w_i \mathbf{a}_i, \quad \text{with } w_i = \frac{s_i - \bar{s}}{\|\mathbf{s} - \bar{s}\|_2}, \tag{24}$$

where $\bar{s}$ is the mean of the scores. This vector σ represents the axis of maximal variation of the head projected into a space aligned with natural language.

B.5.2. Sparse Coding via Orthogonal Matching Pursuit (OMP)

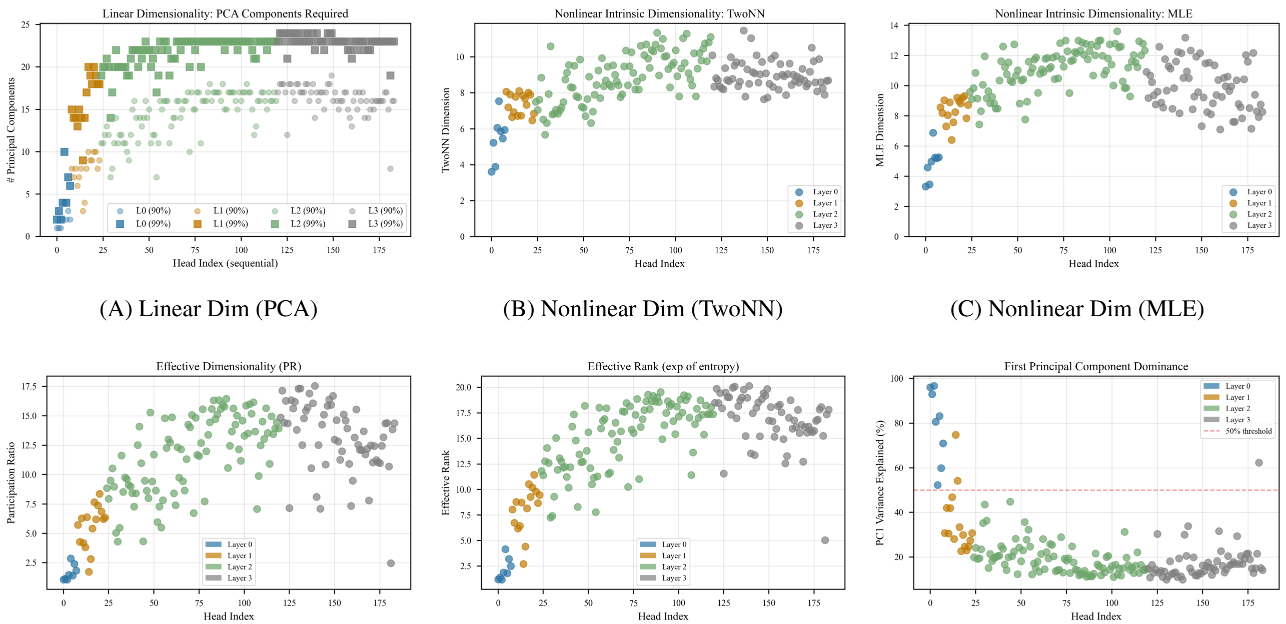
To translate $\sigma_{\ell,b,h}$ into interpretable terms, I use a textual dictionary $\mathcal{D}$ composed of M audio descriptions (e.g. "dog barking", "high pitched tone"). Let $\{\mathbf{t}_j\}_{j=1}^M$ be the corresponding CLAP text embeddings. Since a head may respond to a combination of concepts, I decompose σ via an *Orthogonal Matching Pursuit* (OMP) algorithm. OMP iteratively selects the k atoms from the dictionary that best explain the semantic direction, solving:

$$\min_{\mathbf{x}} \|\sigma_{\ell,b,h} - \mathbf{T}\mathbf{x}\|_2 \quad \text{s.t. } \|\mathbf{x}\|_0 \leq k, \tag{25}$$

where $\mathbf{T}$ is the dictionary embedding matrix. In our experiments I fix $k = 5$. This greedy approach allows us to distinguish between highly specialised heads (pointing to a single atom) and heads that capture broader acoustic concepts.

B.5.3. Metrics for Semantic Coherence

To quantify the quality of head specialization, I use the **Coherence Z-score**. Let $\mathcal{I} = \{j_1, \ldots, j_k\} \subseteq \{1, \ldots, M\}$ be the set of indices of the k atoms selected by OMP for a



(A) Linear Dim (PCA) (B) Nonlinear Dim (TwoNN) (C) Nonlinear Dim (MLE)

(D) Participation Ratio (E) Effective Rank (F) PC1 Dominance

Figure 4. **Granular analysis of the 184 HTS-AT heads.** The panels show the distribution of dimensionality metrics for each head, indexed sequentially. (A–C) Comparison between linear and nonlinear estimates; (D–E) Spectral spread metrics; (F) Variance concentration on the first component.

given head, and let $\{\mathbf{t}_j\}_{j=1}^{M}$ be the ℓ_2-normalised text embeddings of the full dictionary.

I define the **intra-group similarity** as the mean pairwise cosine similarity among the selected atoms[3]:

$$S_{\text{intra}} = \frac{1}{k(k-1)} \sum_{\substack{i,j \in \mathcal{I} \\ i \neq j}} \mathbf{t}_i^\top \mathbf{t}_j. \tag{26}$$

To establish a reference distribution, I compute the similarity of each selected atom against all M dictionary entries. Collecting all $k \cdot M$ such values, I define:

$$\mu_{\mathcal{D}} = \frac{1}{kM} \sum_{i \in \mathcal{I}} \sum_{j=1}^{M} \mathbf{t}_i^\top \mathbf{t}_j, \tag{27}$$

$$\sigma_{\mathcal{D}} = \sqrt{\frac{1}{kM} \sum_{i \in \mathcal{I}} \sum_{j=1}^{M} \left(\mathbf{t}_i^\top \mathbf{t}_j - \mu_{\mathcal{D}}\right)^2}. \tag{28}$$

[3]For each i we have $k-1$ choices for j. Thus the total number of comparisons is $k(k-1)$, as we count (i,j) and (j,i) twice.

The Coherence Z-score[4] is then:

$$Z = \frac{S_{\text{intra}} - \mu_{\mathcal{D}}}{\sigma_{\mathcal{D}} + \varepsilon}, \tag{29}$$

where $\varepsilon = 10^{-8}$ ensures numerical stability.

A large positive Z indicates that the k descriptions selected by OMP are significantly more similar to each other than to the dictionary at large, signalling that the head is focused on a well-defined and internally consistent semantic domain. Conversely, values near zero or negative indicate that the selected atoms span heterogeneous concepts, pointing to weak or spurious specialization.

B.5.4. Background of this Approach

The proposed cross-modal grounding framework draws on three lines of prior work. First, the idea of decomposing audio embeddings into sparse combinations of human-interpretable textual concepts has been explored by Zhang et al. (2025b), who apply sparse linear decomposition over a vocabulary of audio descriptions to make CLAP embeddings interpretable. Second, the analysis of attention head specialization via sparse dictionary matching was in-

[4]Note that $\mu_{\mathcal{D}}$ and $\sigma_{\mathcal{D}}$ are computed over the full set of $k \cdot M$ atom-to-dictionary similarities, providing a background distribution against which the intra-group cohesion S_{intra} is normalised. This differs from a permutation-based Z-score, but offers a closed-form and computationally efficient alternative.

troduced in the vision-language domain by Basile et al. (2025), together with the ResiDual original paper Basile et al. (2024), who use Simultaneous Orthogonal Matching Pursuit over a model's unembedding space to identify heads aligned with specific semantic attributes. Third, in the vision domain, Gandelsman et al. (2023) decompose the output of individual attention heads in CLIP's image encoder and interpret the resulting directions via CLIP's own text representations, revealing property-specific roles such as location or shape.

My approach differs from all three in a key respect: rather than operating within the embedding space of a single model, I use CLAP as an *external* semantic bridge to interpret the internal representations of HTS-AT. This allows us to ground head-level activations in natural language without requiring HTS-AT to natively operate in a shared audio-text space.

B.6. Results: Semantic Landscapes and Head Specialization

The application of the cross-modal grounding framework reveals the emergence of functional specialisation that follows the structural hierarchy of the model. The analysis of the semantic coherence Z-score (Figure 5) and the identification of the most specialised heads (Table 4) shed light on how abstract acoustic concepts crystallise into semantic categories.

B.6.1. Emergence of Semantic Coherence

The evolution of semantic coherence across layers (Figure 5) reveals a phase transition linked to the hierarchical structure of the model:

- **Layers 0 and 1 (Semantic Void):** In the first four blocks (0–3), semantic coherence is predominantly negative or close to zero. At this stage, heads extract low-dimensional features ($L < 19$) with high spectral dominance. The low coherence indicates that these heads respond to physical properties of the signal that are common to multiple sources, making a unique mapping onto specific textual descriptions impossible.
- **Layer 2 (Functional Diversification):** This stage marks the transition into a regime of increased non-linearity. In particular, a clear increase in head specialization emerges toward the end of the layer, in concomitance with the growth of the intrinsic non-linear dimensionality (as measured by TwoNN, Fig. 3) and the concurrent decrease in the leading eigenvalue share (EVR_1), which drops from 0.24 at the beginning of Layer 2 to 0.16 at its end (Fig. 3).

  This joint behaviour suggests that the expansion of the underlying manifold complexity is closely linked to the emergence of more semantically selective attention heads. Accordingly, highly specialized heads begin to appear, such as `L2_B4_H1` ($Z = 1.46$, absolute block 8), which capture fine-grained acoustic events associated with the *human_body* category.
- **Layer 3 (Semantic Convergence):** This layer initially continues the trend observed at the end of Layer 2, with a high density of more specialized heads in the early blocks of Layer 3. In this regime, heads remain well-aligned with CLAP semantic concepts, despite the fact that the leading eigenvalue share (EVR_1) reaches one of its lowest values (0.15), indicating that information is distributed across a larger number of latent components.

  However, this trend progressively reverses toward the final blocks (10–11), where a mild decrease in head specialization is observed. This reversal is consistent with a broader inversion in all structural metrics, including L, N, L/N, and EVR_1, which collectively shift from their previously increasing or decreasing trends (see Fig. 3).

  Overall, Layer 3 appears to exhibit an initial phase of strong semantic alignment, followed by a late-stage relaxation of specialization, suggesting a partial re-mixing of representations in the deepest blocks.

B.6.2. Identification of Specialized Heads

Table 4 reports a selection of the heads showing the highest semantic coherence, revealing a preference for categories with strong timbral or temporal characteristics.

The results reported in Table 4 confirm the emergence of a clear semantic organization of attention heads across layers. In particular, the highest coherence scores are consistently associated with heads that capture well-defined acoustic categories such as *human_body*, *nature*, and *urban sounds*, suggesting that specialization is not random but structurally aligned with meaningful audio concepts.

Finally, the presence of moderately coherent but still structured heads in later blocks suggests that, while full specialization emerges primarily in intermediate layers, the deepest representations undergo a partial re-mixing of features, consistent with the non-monotonic behaviour observed in Fig. 5 and Fig. 3. Importantly, the observed inversion of the trend in the deepest blocks is also consistent with the findings reported in (Basile et al., 2024), indicating a shared structural behaviour across architectures and suggesting that this non-monotonic specialization pattern extends beyond the audio domain to image-domain residual stream models as well.

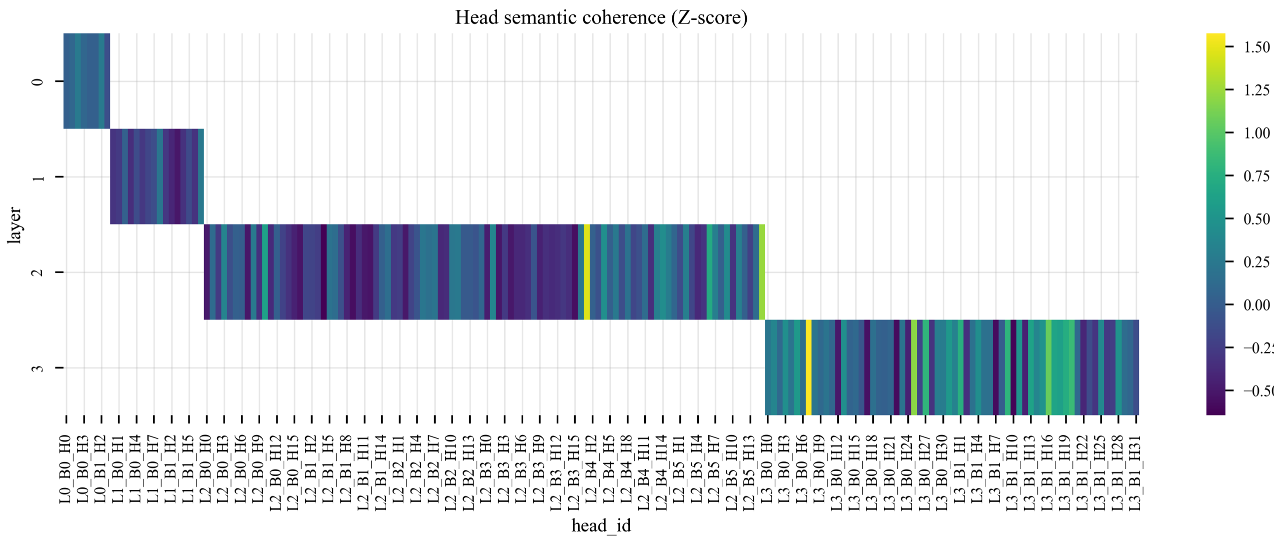


Figure 5. **Heatmap of the semantic coherence Z-score.** The map highlights how the capacity for semantic alignment emerges from the end of Layer 2 onward, coinciding with the increase in nonlinear dimensionality and the reduction in the dominance of the first principal component.

C. Reweighting Pipeline

This section describes the full implementation of ResiDual* (RD*) integrated into the frozen HTS-AT transfer of CLAP model. The pipeline consists of two sequential phases: a *fitting pass* that estimates a fixed PCA basis for each attention head (§C.2), followed by an *optimisation phase* in which the per-head weight vectors are refined as free parameters via gradient descent to maximise zero-shot classification accuracy (§C.3). The forward-pass reweighting applied at inference time is described in §C.4.

Throughout this section I assume batch size $B = 1$ for notational clarity; all operations extend straightforwardly to $B > 1$ by treating the batch axis independently. I use the index conventions of Appendix A: $\ell \in \{0, 1, 2, 3\}$ is the stage index, $b \in \{1, \ldots, B_\ell\}$ is the block index within stage ℓ, and $h \in \{1, \ldots, H_\ell\}$ is the head index. The per-head feature dimension is $d_h = 24$ (constant across all stages), $N_w^\ell = S_\ell^2/M$ is the number of attention windows at stage ℓ, and $M = 64$ is the number of tokens per window. All symbols are collected in Table 3.

C.1. Architecture and Module Injection

C.1.1. Spectral reweighting layers.

For each target stage $\ell \in \mathcal{L}$, block b, and head h, RD* instantiates a `SpectralReweightingLayer` — a lightweight `nn.Module` that carries two non-trainable buffers, $\mathbf{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$ and $\boldsymbol{\mu}_{\ell,b,h} \in \mathbb{R}^{d_h}$, and a single learnable parameter vector $\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$. All such modules are collected in a nested `nn.ModuleDict` indexed by (ℓ, b, h), constructed once during model initialisation in `_build_spectral_layers` from the stage dimensions and head counts of the underlying HTS-AT configuration; they persist in the model's `state_dict` across saves and loads. Each module exposes two methods: `fit_pca`, which populates $\mathbf{\Phi}_{\ell,b,h}$ and $\boldsymbol{\mu}_{\ell,b,h}$ from data and initialises $\boldsymbol{\lambda}_{\ell,b,h}$ (§C.2); and `forward`, which applies the RD* transform (§C.4).

Figure 6 summarises the full lifecycle of a `SpectralReweightingLayer`: the PCA fitting pass accumulates N_{clips} training samples to derive $\boldsymbol{\mu}_{\ell,b,h}$, $\mathbf{\Phi}_{\ell,b,h}$, and the initial weight vector $\boldsymbol{\lambda}_{\ell,b,h}^{(0)}$ (Eqs. (35)–(36)); the optimisation phase then refines $\boldsymbol{\lambda}_{\ell,b,h}$ by minimising $\mathcal{J}(\boldsymbol{\lambda})$ (Eq. (37)); at inference time the four-step spectral reweighting transform (Eqs. (39)–(44)) is applied to each incoming $\mathbf{H}_{\ell,b,h}$.

C.1.2. Module injection via forward hooks.

RD* injects its reweighting through permanent PyTorch forward hooks. For each target block (ℓ, b) with $\ell \in \mathcal{L}$, a dedicated `AttentionHook` object is registered once on the corresponding `WindowAttention` module at model construction via `register_forward_hook` and remains attached throughout the entire model lifetime — no registration or removal occurs at inference time. The hook is stateless: it holds only a reference to the corresponding slice `spectral_layers`$[\ell][b]$ of the module dict, and operates in one of two modes controlled by a `collect_for_fitting` flag. In *collec-*

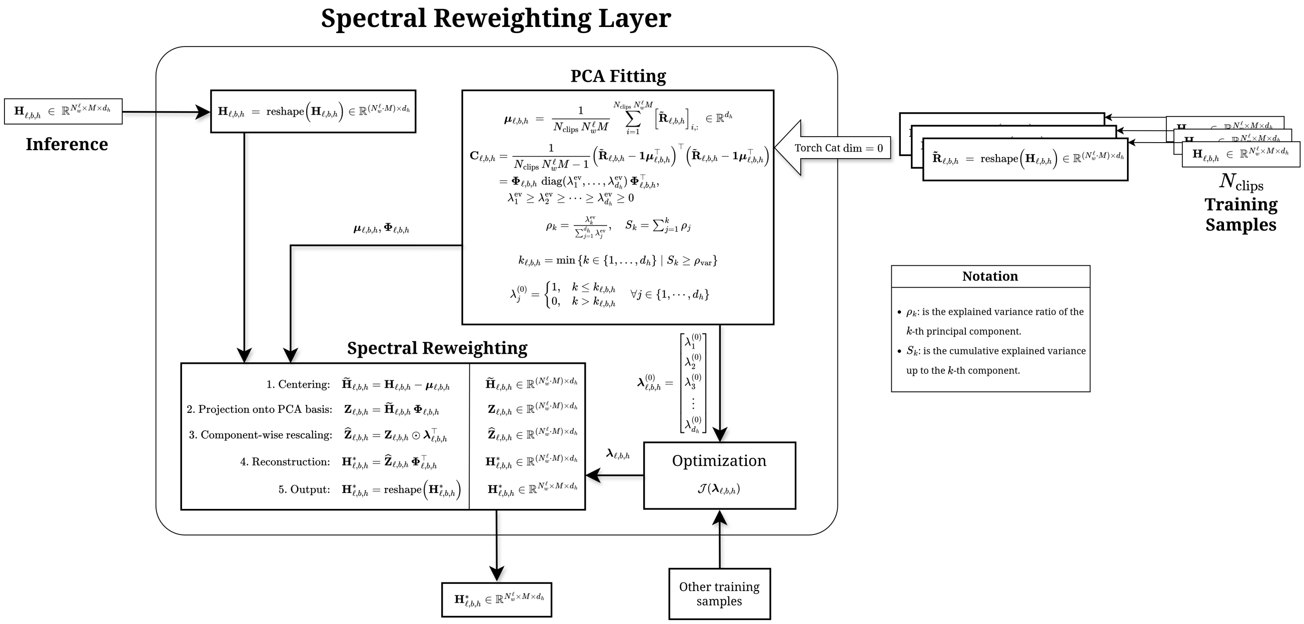


Figure 6. Internal structure of a `SpectralReweightingLayer` for head h of block (ℓ, b). **Right** (*fitting pass*): N_{clips} per-head activation matrices are concatenated to form the fitting matrix $\widetilde{\mathbf{R}}_{\ell,b,h}$ (Eq. (30)), from which the empirical mean $\boldsymbol{\mu}_{\ell,b,h}$ and PCA basis $\boldsymbol{\Phi}_{\ell,b,h}$ are estimated (Eq. (32)), and the initial weight vector $\boldsymbol{\lambda}^{(0)}_{\ell,b,h}$ is set via the variance-threshold criterion (Eqs. (35)–(36)). **Centre** (*spectral reweighting*): at inference time the four-step transform centres, projects onto the PCA basis, rescales per component, and reconstructs each head tensor (Eqs. (39)–(44)). **Bottom-right** (*optimisation*): the weight vector $\boldsymbol{\lambda}_{\ell,b,h}$ is refined by minimising $\mathcal{J}$ on additional training samples (§C.3).

tion mode, it accumulates the raw per-head tensors $\mathbf{H}_{\ell,b,h}$ without modifying the forward-pass output, feeding the fitting pass of §C.2. In *reweighting mode*, it replaces $\mathbf{H}_{\ell,b,h}$ with the reweighted output $\mathbf{H}^*_{\ell,b,h}$ (Eq. (42)) and immediately re-applies the output projection $W^O_{\ell,b}$ and dropout `proj_drop` within the hook itself before returning the modified output[5]; this is necessary because returning a modified tensor from the hook bypasses the projection that would otherwise follow in the original `WindowAttention` forward. No structural modification of the original model is performed: all pretrained parameters — the fused QKV projection $W^{QKV}_{\ell,b}$, the output projection $W^O_{\ell,b}$ with bias $\mathbf{b}^O_{\ell,b}$, and the relative position bias table $\mathbf{B}_{\ell,b,h}$ — remain exactly as loaded from the checkpoint.

Figure 7 illustrates the resulting data flow for a single target block (ℓ, b): the `WindowAttention` module produces one tensor $\mathbf{H}_{\ell,b,h}$ per head, each of which is intercepted by the corresponding hook and processed independently by the `SpectralReweightingLayer` before the outputs are reassembled and passed to the output projection $W^O_{\ell,b}$.

[5]Since the hook receives the output of `WindowAttention` rather than the intermediate pre-projection activations, the per-head tensors $\mathbf{H}_{\ell,b,h}$ are recovered by re-running the fused QKV projection $W^{QKV}_{\ell,b}$ on the hook's `inputs[0]` and recomputing $\mathbf{H}_{\ell,b,h} = \boldsymbol{\alpha}_{\ell,b,h}\mathbf{V}_{\ell,b,h}$ within the hook itself. This incurs a second QKV computation per forward pass at the target blocks, which is negligible given that $\boldsymbol{\Phi}_{\ell,b,h}$ and $\boldsymbol{\lambda}_{\ell,b,h}$ operate in the low-dimensional space $\mathbb{R}^{d_h}$ with $d_h = 24$.

C.2. Fitting Pass: PCA Basis Estimation

C.2.1. Data collection.

The fitting pass is run with gradients disabled and the model in evaluation mode. For each audio clip (up to $N_{\max}$ clips), a forward pass is executed with data collection enabled. The hook registered at block (ℓ, b) intercepts, for every head h, the tensor $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$ and flattens it along the spatial axes to obtain a matrix of shape $\mathbb{R}^{(N_w^\ell \cdot M) \times d_h}$, which is appended to an accumulation buffer for the triplet (ℓ, b, h). After all clips have been processed, the per-clip contributions are stacked to form the *fitting matrix*

$$\widetilde{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{N_{\text{fit}} \times d_h}, \qquad N_{\text{fit}} = N_{\text{clips}} \cdot N_w^\ell \cdot M, \quad (30)$$

where $N_{\text{clips}} \le N_{\max}$ is the number of processed clips.

C.2.2. PCA decomposition.

Standard PCA is applied to $\widetilde{\mathbf{R}}_{\ell,b,h}$. The empirical mean

$$\boldsymbol{\mu}_{\ell,b,h} = \frac{1}{N_{\text{fit}}} \sum_{i=1}^{N_{\text{fit}}} \widetilde{\mathbf{R}}_{\ell,b,h}[i,:] \in \mathbb{R}^{d_h} \quad (31)$$

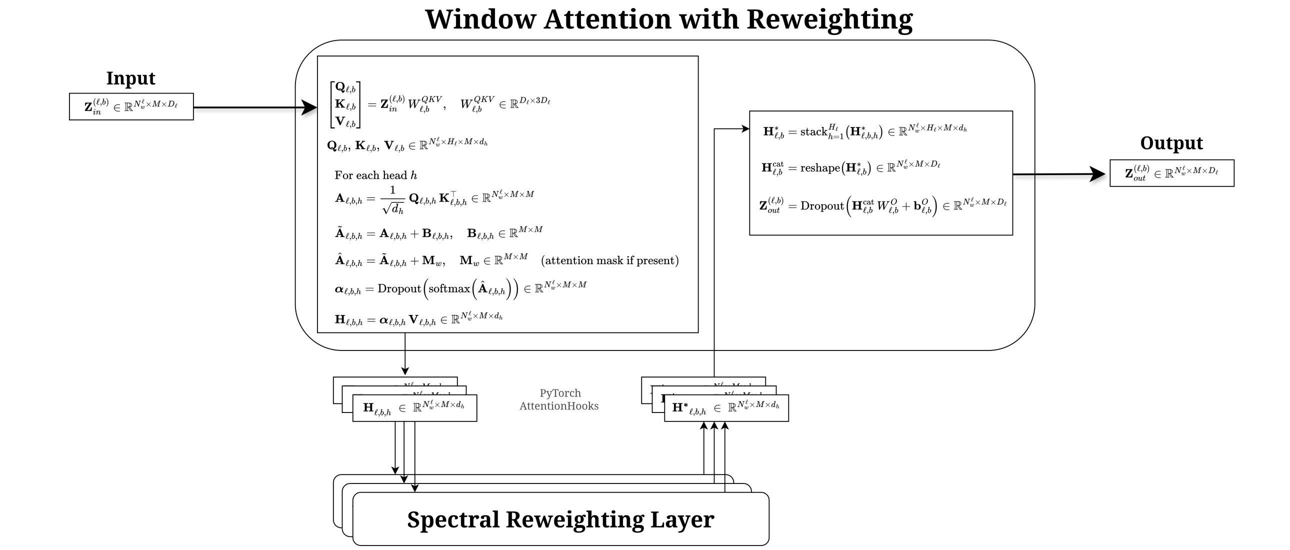


Figure 7. Reweighting pipeline for a single target block (ℓ, b). The `WindowAttention` module computes the raw per-head tensors $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$ via the standard scaled dot-product attention with relative position bias $\mathbf{B}_{\ell,b,h}$. PyTorch `AttentionHooks` intercept each $\mathbf{H}_{\ell,b,h}$ and route it through the corresponding `SpectralReweightingLayer`; the reweighted outputs $\mathbf{H}^*_{\ell,b,h}$ are then stacked, reshaped, and projected by $W^O_{\ell,b}$ to yield the block output $\mathbf{Z}^{(\ell,b)}_{\text{out}} \in \mathbb{R}^{N_w^\ell \times M \times D_\ell}$.

is computed and subtracted, and the centred empirical covariance is decomposed as

$$\mathbf{C}_{\ell,b,h} = \frac{1}{N_{\text{fit}} - 1}\Big(\widetilde{\mathbf{R}}_{\ell,b,h} - \mathbf{1}\boldsymbol{\mu}_{\ell,b,h}^\top\Big)^\top \Big(\widetilde{\mathbf{R}}_{\ell,b,h} - \mathbf{1}\boldsymbol{\mu}_{\ell,b,h}^\top\Big)$$

$$= \boldsymbol{\Phi}_{\ell,b,h} \, \operatorname{diag}(\lambda_1^{\text{ev}}, \ldots, \lambda_{d_h}^{\text{ev}}) \, \boldsymbol{\Phi}_{\ell,b,h}^\top, \tag{32}$$

$$\text{with } \lambda_1^{\text{ev}} \geq \cdots \geq \lambda_{d_h}^{\text{ev}} \geq 0, \tag{33}$$

where $\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$ is the orthogonal matrix of eigenvectors (columns ordered by decreasing eigenvalue) and λ_j^{ev} denotes the j-th eigenvalue of $\mathbf{C}_{\ell,b,h}$. I use the superscript "ev" throughout to distinguish these fixed quantities from the trainable weight vector $\boldsymbol{\lambda}_{\ell,b,h}$. Both $\boldsymbol{\Phi}_{\ell,b,h}$ and $\boldsymbol{\mu}_{\ell,b,h}$ are stored as non-trainable buffers; the accumulation buffers $\widetilde{\mathbf{R}}_{\ell,b,h}$ are then cleared to free memory.

C.2.3. Selection of the task-relevant subspace.

The number of principal components retained for each head is determined by a variance-threshold criterion. Define the normalised eigenvalue shares and their cumulative sum as

$$\rho_k = \frac{\lambda_k^{\text{ev}}}{\sum_{j=1}^{d_h} \lambda_j^{\text{ev}}}, \qquad S_k = \sum_{j=1}^{k} \rho_j, \tag{34}$$

so that $S_{d_h} = 1$ by construction. The retained dimensionality is then the smallest k sufficient to explain a fraction ρ_{var} of the total variance:

$$k_{\ell,b,h} = \min\{k \in \{1, \ldots, d_h\} \mid S_k \geq \rho_{\text{var}}\}. \tag{35}$$

Because $d_h = 24$ is small and constant, this threshold is evaluated exactly over all components without approximation. The default value $\rho_{\text{var}} = 0.95$ is used throughout.

C.2.4. Initialisation of $\boldsymbol{\lambda}_{\ell,b,h}$.

For each head (ℓ, b, h), the index $k_{\ell,b,h}$ defined in Eq. (35) identifies the smallest number of principal components that together explain at least a fraction ρ_{var} of the total variance, providing a per-head estimate of the effective dimensionality of the task-relevant subspace. The learnable weight vector $\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$ is initialised component-wise as

$$\lambda_j^{(0)} = \begin{cases} 1 & j \leq k_{\ell,b,h}, \\ 0 & j > k_{\ell,b,h}, \end{cases} \tag{36}$$

corresponding to the identity map restricted to the task-relevant subspace and complete suppression of the remaining noise components. At this initialisation point, RD* therefore acts as a projection-and-reconstruction operator that retains the top-$k_{\ell,b,h}$ PCA directions unchanged and zeros out the remainder.

C.3. Optimisation of $\boldsymbol{\lambda}$ via Zero-Shot Accuracy

After the fitting pass has fixed the PCA bases $\{\boldsymbol{\Phi}_{\ell,b,h}\}$ and initialised $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ via Eq. (36), the second phase optimises the weight vectors to maximise zero-shot classification accuracy on a held-out validation set.

C.3.1. PARAMETER ISOLATION.

All model parameters except the weight vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}_{\ell\in\mathcal{L},\,b,\,h}$ are frozen by setting `requires_grad = False`. The only degrees of freedom during optimisation are therefore the $\sum_{\ell\in\mathcal{L}} B_\ell \cdot H_\ell$ weight vectors, each of dimension $d_h = 24$, that control the per-component gain of each attention head in the target stages $\mathcal{L}$.

C.3.2. OBJECTIVE.

Given a set of audio clips with ground-truth class labels $y \in \{1, \dots, C\}$ and pre-computed ℓ_2-normalised text embeddings $\mathbf{t}_c \in \mathbb{R}^{d_{\text{proj}}}$ for each class c, the optimisation objective is the cross-entropy loss over zero-shot logits:

$$\mathcal{J}(\boldsymbol{\lambda}) = -\mathbb{E}_{(x,\,y)}\left[\log \frac{\exp\big(\hat{\mathbf{a}}(x)^\top \mathbf{t}_y\big)}{\sum_{c=1}^{C} \exp\big(\hat{\mathbf{a}}(x)^\top \mathbf{t}_c\big)}\right], \tag{37}$$

where $\hat{\mathbf{a}}(x) \in \mathbb{R}^{d_{\text{proj}}}$ is the ℓ_2-normalised audio embedding produced by P applied to the spatially averaged Stage-3 output of HTS-AT after the RD* transforms have been applied (cf. Table 3). Minimising $\mathcal{J}$ is equivalent to maximising zero-shot classification accuracy in expectation.

C.3.3. OPTIMISER AND SCHEDULE.

The weight vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ are updated with AdamW (Loshchilov & Hutter, 2017) at learning rate η and weight decay γ; all hyperparameter values are reported in Table 3. The CLAP audio projection head $P : \mathbb{R}^{D_3} \to \mathbb{R}^{d_{\text{proj}}}$ is kept frozen but is included in the computational graph so that gradients flow from the similarity space back to $\boldsymbol{\lambda}_{\ell,b,h}$.

C.3.4. EARLY STOPPING AND BEST-$\boldsymbol{\lambda}$ RESTORATION.

At the end of each epoch, zero-shot accuracy on the validation set is evaluated. A snapshot of the current $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ is saved whenever a new accuracy maximum is achieved. Training halts early if no improvement is observed for p consecutive epochs; at termination, the best snapshot is restored, guaranteeing that the final weight vectors correspond to the configuration achieving the highest observed validation accuracy.

C.3.5. HYPERPARAMETER SUMMARY.

Table 3 collects all hyperparameters introduced by RD* together with the default values used in all experiments.

C.4. Inference: Forward Pass with Spectral Reweighting

At inference time, the hook at block (ℓ, b) intercepts the raw head outputs and applies the following sequence of operations independently for each head h.

Step 0 — Flattening. The raw head output, originally shaped as $\mathbb{R}^{N_w^\ell \times M \times d_h}$, is first flattened along the first two axes:

$$\mathbf{H}_{\ell,b,h} = \texttt{reshape}(\mathbf{H}_{\ell,b,h}) \in \mathbb{R}^{(N_w^\ell \cdot M)\times d_h}, \tag{38}$$

so that all subsequent operations act on a 2-D matrix of token representations.

Step 1 — Centering.

$$\widetilde{\mathbf{H}}_{\ell,b,h} = \mathbf{H}_{\ell,b,h} - \boldsymbol{\mu}_{\ell,b,h} \in \mathbb{R}^{(N_w^\ell \cdot M)\times d_h}, \tag{39}$$

where $\boldsymbol{\mu}_{\ell,b,h} \in \mathbb{R}^{d_h}$ is broadcast over the first axis.

Step 2 — Projection onto the full PCA basis.

$$\mathbf{Z}_{\ell,b,h} = \widetilde{\mathbf{H}}_{\ell,b,h}\,\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{(N_w^\ell \cdot M)\times d_h}. \tag{40}$$

Unlike fixed-weight variants that truncate to $k_{\ell,b,h} < d_h$ components, RD* projects onto the *full* basis $\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h\times d_h}$, delegating effective truncation to the learned weights.

Step 3 — Per-component rescaling.

$$\widehat{\mathbf{Z}}_{\ell,b,h} = \mathbf{Z}_{\ell,b,h} \odot \boldsymbol{\lambda}_{\ell,b,h}^\top \in \mathbb{R}^{(N_w^\ell \cdot M)\times d_h}, \tag{41}$$

where $\boldsymbol{\lambda}_{\ell,b,h}^\top \in \mathbb{R}^{d_h}$ is broadcast over the first axis. Each entry λ_j can amplify ($|\lambda_j| > 1$), attenuate ($|\lambda_j| \in (0,1)$), suppress ($\lambda_j = 0$), or invert ($\lambda_j < 0$) the corresponding principal component.

Step 4 — Reconstruction in the original space.

$$\mathbf{H}^*_{\ell,b,h} = \widehat{\mathbf{Z}}_{\ell,b,h}\,\boldsymbol{\Phi}_{\ell,b,h}^\top \in \mathbb{R}^{(N_w^\ell \cdot M)\times d_h}. \tag{42}$$

The empirical mean $\boldsymbol{\mu}_{\ell,b,h}$ is not restored after reconstruction, following the original ResiDual formulation (Basile et al., 2024). In this setting, the mean represents the average constant contribution of head h to the residual stream, which is suppressed by design to retain only the discriminative, variance-carrying directions. Any residual shift introduced by this omission is implicitly compensated during the optimisation of $\boldsymbol{\lambda}_{\ell,b,h}$, since the weight vectors are free to adjust for a constant offset in the PCA-projected space. (cf. Eq. (12)).

Step 5 — Reshaping. Before being passed to the output projection, $\mathbf{H}^*_{\ell,b,h}$ is reshaped back to the original spatial layout:

$$\mathbf{H}^*_{\ell,b,h} \leftarrow \texttt{reshape}(\mathbf{H}^*_{\ell,b,h}) \in \mathbb{R}^{N_w^\ell \times M \times d_h}. \tag{43}$$

Combining Steps 1–4, the full RD* transform (on the flattened representation) reads:

$$\begin{aligned} \mathrm{RD}^*(\mathbf{H}_{\ell,b,h},\, \boldsymbol{\lambda}_{\ell,b,h}) \;\in\; \mathbb{R}^{(N_w^\ell \cdot M)\times d_h} \;= \\ = \big(\mathbf{H}_{\ell,b,h} - \boldsymbol{\mu}_{\ell,b,h}\big)\, \boldsymbol{\Phi}_{\ell,b,h}\; \mathrm{diag}(\boldsymbol{\lambda}_{\ell,b,h})\, \boldsymbol{\Phi}_{\ell,b,h}^\top. \end{aligned} \tag{44}$$

The reshaped $\mathbf{H}^*_{\ell,b,h}$ replaces $\mathbf{H}_{\ell,b,h}$ in Eq. (12), so the output projection $W^O_{\ell,b}$ and the residual-stream update of Eq. (8) proceed unchanged. During optimisation, gradients of any downstream loss flow through RD^* and update only $\boldsymbol{\lambda}_{\ell,b,h}$; the basis $\boldsymbol{\Phi}_{\ell,b,h}$, the mean $\boldsymbol{\mu}_{\ell,b,h}$, and all original model parameters are held fixed.

D. K-Fold Evaluation Pipeline and Results

D.1. Experimental Protocol

To obtain reliable and unbiased estimates of zero-shot classification accuracy, we adopt a stratified K-fold cross-validation protocol ($K = 5$) applied uniformly to both the CLAP baseline and ResiDual*. All experiments are conducted on four publicly available audio classification benchmarks: ESC-50 (Piczak), IRMAS (Juan J. Bosch & Herrera), TinySOL (Emanuele et al., 2020), and VocalSound (Gong et al., 2022), which collectively span environmental sounds, musical instruments, orchestral timbres, and human vocal categories.

D.1.1. Dataset sampling.

For each dataset, up to $N_{\text{samples}} = 2{,}000$ examples are selected via class-stratified sampling with a fixed random seed, ensuring a balanced class distribution across all folds. The stratified sampling procedure first computes the per-class allocation $\lfloor N_{\text{samples}}/C \rfloor$ (plus a unit remainder distributed to the first $N_{\text{samples}} \bmod C$ classes), then samples without replacement from each class independently, guaranteeing reproducibility via a global seed.

D.1.2. Cross-validation split.

The sampled subset is partitioned into $K = 5$ stratified folds using `StratifiedKFold` with `shuffle=True` and a fixed `random_state`. At each fold iteration, the training split (four folds) is further divided into two non-overlapping sub-partitions: a *PCA fitting* partition of $N_{\text{PCA}} = 250$ samples, used exclusively to estimate the spectral bases $\{\boldsymbol{\Phi}_{\ell,b,h},\, \boldsymbol{\mu}_{\ell,b,h}\}$ (cf. §C.2), and a λ-*optimisation* partition comprising the remaining training examples, used to refine the weight vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ via the objective of Eq. (37). The held-out fold (one fifth of the sampled subset) is reserved exclusively for final zero-shot evaluation.

Figure 8 illustrates the resulting data partitioning scheme across all five folds, highlighting the PCA, λ-optimisation, and final validation splits.

D.1.3. Text prompt construction.

Class-conditional text embeddings are computed once per dataset prior to the fold loop. Each class label c is formatted as the natural-language prompt `"this is the sound of {c}"`, where underscores are replaced by spaces. This prompt template follows the zero-shot inference convention adopted in the official Microsoft CLAP repository (Elizalde et al., 2022), where the same phrasing is used for audio classification evaluation across multiple benchmarks. The resulting prompts are encoded by the frozen CLAP text encoder and ℓ_2-normalised to obtain $\mathbf{t}_c \in \mathbb{R}^{d_{\text{proj}}}$ for each class $c \in \{1, \dots, C\}$.

D.1.4. Zero-shot inference.

At evaluation time, the audio embedding $\hat{\mathbf{a}}(x)$ for a test clip x is obtained by passing the raw waveform through the (reweighted) audio encoder followed by the projection head P, and ℓ_2-normalising the result. The predicted class is then

$$\hat{y}(x) \;=\; \arg\max_{c \in \{1,\dots,C\}} \hat{\mathbf{a}}(x)^\top \mathbf{t}_c. \tag{45}$$

For the CLAP baseline, no adaptation is performed and the frozen model is evaluated directly. For ResiDual*, the per-fold PCA fitting and λ-optimisation phases precede evaluation, and the model weights $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ are fully reset between folds by reloading the CLAP checkpoint and re-instantiating all `SpectralReweightingLayer` modules.

D.1.5. Optimiser configuration.

The weight vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ are updated with AdamW (Loshchilov & Hutter, 2017) with learning rate $\eta = 5 \times 10^{-3}$ and weight decay $\gamma = 5 \times 10^{-2}$, for a maximum of $T_{\max} = 40$ epochs with early stopping patience $p = 5$. The target stages are $\mathcal{L} = \{2, 3\}$ and the PCA variance threshold is $\rho_{\text{var}} = 0.95$. All other hyperparameters follow Table 3.

D.2. Quantitative Results

Table 5 reports the mean zero-shot accuracy and standard deviation across the five folds for each dataset and each model variant, comparing three ResiDual* configurations that differ only in the set of target stages $\mathcal{L}$: reweighting applied to all stages ($\mathcal{L} = \{0, 1, 2, 3\}$), to the two deepest stages only ($\mathcal{L} = \{2, 3\}$, default), and to Stage 3 alone ($\mathcal{L} = \{3\}$). Bold entries indicate the highest mean accuracy for each dataset across all variants. ResiDual* with $\mathcal{L} = \{2, 3\}$ improves over the CLAP baseline on all four benchmarks, with gains ranging from $+2.4$ percentage points on ESC-50 to $+54.8$ percentage points on TinySOL.

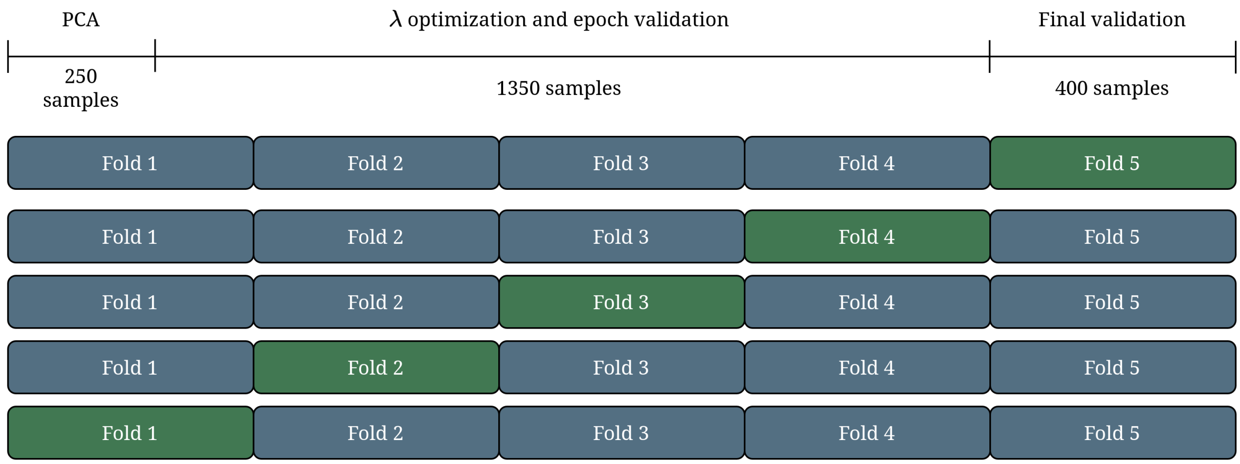


Figure 8. Cross-validation partitioning scheme. The dataset is split into $K = 5$ stratified folds: the first 250 samples of each training split are reserved for PCA fitting (left bracket), the remaining training examples are used for λ-optimisation (centre), and the held-out fold (green) serves as the final validation set.

D.2.1. Effect of target stage selection.

The comparison across the three configurations in Table 5 reveals a clear and consistent pattern: restricting reweighting to the two deepest stages ($\mathcal{L} = \{2, 3\}$) yields the best results on all four benchmarks, while extending it to all stages ($\mathcal{L} = \{0, 1, 2, 3\}$) or restricting it to Stage 3 alone ($\mathcal{L} = \{3\}$) both lead to lower accuracy.

The underperformance of $\mathcal{L} = \{0, 1, 2, 3\}$ relative to the default $\mathcal{L} = \{2, 3\}$ is particularly informative. As established by the head analysis in Appendix B (Figures 3–4), early-stage heads (Stages 0 and 1) operate in near-unidimensional regimes with very high PC1 dominance ($\mathrm{EVR}_1 \approx 0.92$ in Stage 0) and near-zero semantic coherence Z-scores, indicating that they encode low-level acoustic properties — such as broadband energy and dominant spectral envelope — that are common across sound categories rather than discriminative for them. Applying spectral reweighting to these heads therefore introduces unnecessary degrees of freedom into the optimisation, risking interference with generic low-level feature extraction that the downstream stages rely upon, without providing additional discriminative signal. The result is a modest but consistent degradation relative to reweighting only Stages 2 and 3, where semantic specialisation has already emerged.

The configuration $\mathcal{L} = \{3\}$ provides a useful ablation showing that Stage 2 contributes meaningfully to the adaptation gain. Stage 2 is precisely the regime identified in Appendix B as the onset of functional diversification, where intrinsic nonlinear dimensionality peaks (L/N ratio at its maximum of 2.65) and the first semantically coherent heads appear (e.g. `L2_B4_H1`, $Z = 1.46$; `L2_B5_H15`, $Z = 1.23$). Excluding Stage 2 from the reweighting thus forfeits the ability to amplify these early-emerging discriminative directions, with the loss being especially pronounced on TinySOL (-16.97 pp relative to $\mathcal{L} = \{2, 3\}$), the benchmark most reliant on fine-grained timbral discrimination.

Taken together, these results validate the stage-selection criterion proposed in Method §3: targeting the stages where semantic coherence is highest and nonlinear manifold complexity is maximal, as identified by the head analysis, is both necessary and sufficient for optimal performance.

Table 6 provides the per-fold breakdown for the default configuration $\mathcal{L} = \{2, 3\}$, offering a more granular view of stability across splits.

D.3. Visualisation and Discussion

Figures 9–11 provide complementary perspectives on the evaluation results.

D.3.1. Aggregate accuracy (Figure 9).

Figure 9 summarises the mean zero-shot accuracy and fold-level variability (shown as ± 1 standard deviation error bars) for both model variants across the four benchmarks. The improvements conferred by ResiDual* are consistent and statistically robust, with the gain being most pronounced on TinySOL ($+54.8$ pp) and IRMAS ($+21.2$ pp) — the two datasets where the CLAP baseline suffers most, likely due to a structural mismatch between the fine-grained timbral distinctions required by these tasks and the generic audio-language alignment captured by the frozen CLAP model. On the semantically richer ESC-50 and VocalSound benchmarks, where CLAP already achieves solid baseline performance, ResiDual* continues to improve results, suggesting that spectral reweighting is additive even in near-saturating regimes.

D.3.2. Per-fold stability (Figure 10).

Figure 10 traces the per-fold accuracy trajectory for each dataset. For both model variants the fold-to-fold variance is low (standard deviation ≤ 1.77 pp in all cases), confirming that neither the baseline nor the adapted model exhibits systematic sensitivity to the particular fold partition. The consistently higher ResiDual* trajectory across all folds and all datasets rules out the possibility that the reported gains arise from favourable fold splits.

D.3.3. Lambda optimisation dynamics (Figure 11).

Figure 11 shows the per-epoch validation accuracy on the λ-optimisation partition, averaged over the five folds (solid line), with individual fold trajectories overlaid in transparency and a ± 1 standard deviation band. The dashed vertical line and annotated dot mark the average best epoch across folds.

Several consistent patterns emerge across datasets. First, the optimisation trajectories exhibit a broadly monotone increasing trend in the early epochs, though with non-negligible plateau phases and occasional minor regressions in later epochs — a behaviour consistent with AdamW on a low-dimensional, non-convex loss landscape. Second, the average best epoch varies substantially across benchmarks: ESC-50 converges earliest (mean best epoch ≈ 21, though with high inter-fold variance ranging from epoch 11 to epoch 32), while IRMAS, TinySOL, and VocalSound all require a larger number of gradient steps (mean best epochs ≈ 37, 38, and 33 respectively). This pattern correlates with the absolute accuracy gain: benchmarks where ResiDual* achieves the largest improvement over the CLAP baseline (TinySOL $+54.8$ pp, IRMAS $+21.2$ pp) are precisely those where the optimiser must explore the λ-space more extensively before identifying a high-performing configuration. Third, the inter-fold variance of the optimisation trajectories is lowest for ESC-50 in terms of final accuracy, but highest in terms of convergence epoch (fold best epochs span 11–32), suggesting that the optimisation landscape for ESC-50 has multiple near-equivalent basins reachable at different depths. IRMAS and TinySOL exhibit the opposite profile: the convergence epoch is consistently late across folds (most folds reach or approach the maximum of 40 epochs), indicating a flatter landscape where marginal improvements accumulate slowly but reliably throughout training.

The gap between the best validation accuracy on the λ-optimisation partition and the held-out test accuracy (Table 5) varies considerably across benchmarks and folds. On ESC-50, the gap is consistently small (≤ 2.5 pp across all folds), indicating near-perfect generalisation of the optimised λ configuration to unseen audio within the same distribution. On IRMAS and VocalSound, however, the gap is substantially larger, reaching up to ≈ 11 pp on individual IRMAS folds and ≈ 10 pp on VocalSound fold 3. This discrepancy between optimisation-partition accuracy and held-out test accuracy does not necessarily imply overfitting of the λ weights in the classical sense: the λ-optimisation partition and the test set are drawn from the same dataset but constitute non-overlapping stratified splits, and the gap likely reflects the limited size of each partition ($\approx$1,350 samples) combined with the inherent difficulty of zero-shot transfer for fine-grained timbral tasks such as IRMAS and VocalSound. Importantly, despite this gap, the held-out test accuracy of ResiDual* remains substantially above the CLAP baseline on all benchmarks and all folds, confirming that the spectral reweighting generalises beyond the optimisation set even when the gap is non-negligible.

D.4. Summary

The optimisation dynamics are consistent with the viability of the RD* pipeline: convergence is broadly monotone in the early training phase, and early stopping reliably identifies configurations that substantially outperform the frozen baseline on held-out data. The gap between optimisation-partition and test accuracy is small on ESC-50 but more pronounced on IRMAS and VocalSound, pointing to a generalisation challenge that scales with task difficulty rather than to overfitting of the λ parameters per se. These results collectively validate the design choices described in Appendix C: a fixed PCA basis estimated on a small unlabelled subset, combined with a lightweight per-component rescaling optimised via cross-entropy on a modest labelled partition, suffices to substantially improve zero-shot classification accuracy without modifying any pretrained parameters, even under conditions of imperfect optimisation-to-test generalisation.

E. Limitations and Future Directions

This appendix collects methodological caveats and directions for future work that emerge from the current implementation of RD*.

Mean pooling vs. token-level analysis. The dimensionality and specialisation analyses of Appendix B rely on mean-pooled head representations $\mathbf{r}_{\ell,b,h}$ (Eq. (15)), collapsing intra-sample spatial variance to a single fixed-size vector per clip. This is motivated by computational tractability: constructing the full token-level matrix

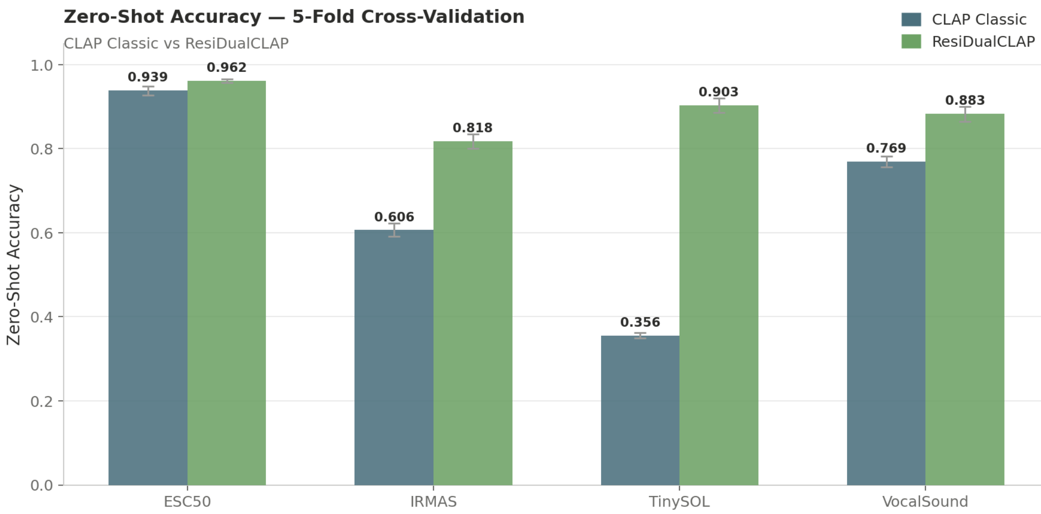


Figure 9. **Mean zero-shot accuracy across benchmarks.** Bar heights correspond to the mean accuracy over the 5 folds; error bars denote ± 1 standard deviation. CLAP Classic (steel blue) vs. ResiDual* (green).

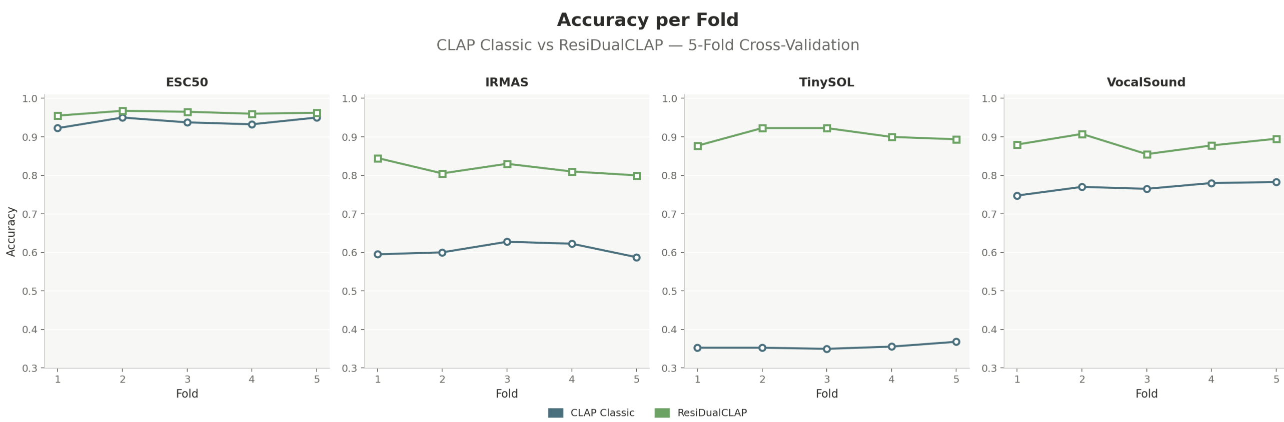


Figure 10. **Fold-by-fold accuracy trajectories.** Each panel corresponds to one benchmark. Circles (CLAP Classic) and squares (ResiDual*) mark individual fold accuracies; lines connect consecutive folds to aid trend visualisation.

$\widetilde{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{n\,N_w^\ell M \times d_h}$ for all $H_{\text{tot}} = 184$ heads simultaneously would exceed available memory. The spectral reweighting operates at full spatial resolution, so this approximation affects only the interpretability analysis. Future work could address it by fitting the PCA basis on a token-level subsample.

Hook placement: pre- vs. post-projection. The current implementation intercepts raw head outputs $\mathbf{H}_{\ell,b,h}$ before the output projection $W^O_{\ell,b}$ (cf. §C.1). A complementary analysis of the post-projection contributions $\widehat{\mathbf{H}}_{\ell,b,h}$ yielded no meaningful differences in geometric structure, motivating the pre-projection formulation for its cleaner cross-stage comparability. A systematic comparison of the two placements under the full RD* optimisation pipeline remains an open empirical question.

Dataset-specific PCA bases. Unlike the original ResiDual (Basile et al., 2024), which transfers a single PCA basis estimated on ImageNet across tasks, RD* estimates $\{\boldsymbol{\Phi}_{\ell,b,h},\ \boldsymbol{\mu}_{\ell,b,h}\}$ from a small task-specific partition ($N_{\text{PCA}} = 250$ samples per fold). Although the original paper shows dataset-specific and ImageNet-based bases to be largely equivalent in isotropic vision transformers (Table 8 of (Basile et al., 2024)), we favour the task-specific strategy for two reasons. First, it avoids potential distribution mismatch: the acoustic heterogeneity of our benchmarks (environmental sounds, orchestral timbres, vocal categories) may not be well covered by any single general-purpose audio corpus. Second, it constitutes a more direct and targeted form of lightweight adaptation, analogous in spirit to fine-tuning but at a fraction of the computational cost.

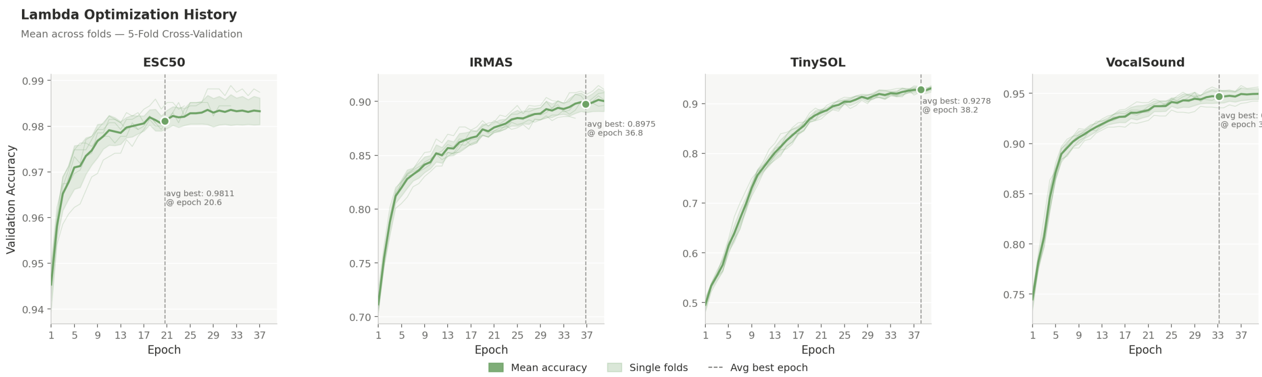


Figure 11. **Lambda optimisation history.** Each panel shows the validation accuracy on the λ-optimisation partition as a function of epoch, for one benchmark. The solid line is the fold-mean; the shaded band is ± 1 standard deviation; semi-transparent lines show individual fold trajectories. The dashed vertical line and annotated marker indicate the average best epoch (where the snapshot used for evaluation was taken).

Scope of the reweighting intervention. ResiDual (Basile et al., 2024) applies spectral reweighting to all residual units, including MLP layers and, in some configurations, the output encoding (RD_Y). RD* intervenes exclusively on attention heads, reflecting the primary objective of this work: to characterise the internal geometry of HTS-AT heads and leverage it for targeted adaptation. Head-level intervention acts directly on the components analysed in Appendix B, avoiding the more entangled representations of MLP layers. Early experiments considered reweighting entire blocks and MLP sub-layers, but head-level intervention was preferred for its greater granularity and its direct correspondence with the geometric analysis. Extending RD* to other residual components remains a natural direction for future work.

Incomplete stage-selection search. Table 5 evaluates only three configurations ($\mathcal{L} \in \{\{3\}, \{2, 3\}, \{0, 1, 2, 3\}\}$); combinations such as $\{1, 2\}$ or $\{1, 2, 3\}$ remain untested, and no formal statistical testing has been conducted. The reported results should therefore be interpreted as indicative rather than conclusive. A full combinatorial search with appropriate significance tests would be needed to establish the optimality of $\mathcal{L} = \{2, 3\}$ with statistical rigour.

Extending the specialisation analysis. The cross-modal grounding of §B.5 identifies each head's dominant semantic axis via its first principal component alone. Jointly considering the top-k components would provide a richer characterisation of heads encoding multiple overlapping concepts. Additionally, the qualitatively noted correlation between intrinsic dimensionality and semantic coherence Z-score has not been formally quantified; doing so would strengthen the theoretical grounding of the stage-selection criterion.

Computational overhead. The permanent forward hooks and per-block QKV re-computation (cf. footnote in §C.1) introduce a modest inference overhead at target blocks. A detailed latency profiling relative to the frozen CLAP baseline has not been included and would be a useful addition for deployment-oriented applications.

Table 3. Summary of notation used throughout the paper. Symbols introduced by RD* are marked with $(\star)$.

Symbol	**Definition**
Architecture and indexing	
$\ell \in \{0,1,2,3\}$	Stage index
$b \in \{1,\ldots,B_\ell\}$	Block index within stage ℓ; $B_\ell \in \{2,2,6,2\}$
$h \in \{1,\ldots,H_\ell\}$	Attention head index; $H_\ell \in \{4,8,16,32\}$
$H_{\text{tot}} = 184$	Total attention heads in HTS-AT $(2\cdot4+2\cdot8+6\cdot16+2\cdot32)$
$S_\ell \in \{64,32,16,8\}$	Spatial side length of the token grid at stage ℓ
$w=8, M=w^2=64$	Attention window side length; tokens per window
$N_w^\ell = S_\ell^2/M \in \{64,16,4,1\}$	Number of attention windows at stage ℓ
$d_h = 24$	Per-head feature dimension (constant); $d_h = D_\ell/H_\ell$
$D_\ell = H_\ell \cdot d_h \in \{96,192,384,768\}$	Total embedding dimension at stage ℓ
$d_{\text{proj}} = 1024$	CLAP joint embedding dimension
Model parameters	
$W_{\ell,b}^{QKV} \in \mathbb{R}^{D_\ell \times 3D_\ell}$	Fused QKV projection at block (ℓ,b)
$W_{\ell,b}^{O} \in \mathbb{R}^{D_\ell \times D_\ell}$	Output projection at block (ℓ,b)
$W_{\ell,b,h}^{O} \in \mathbb{R}^{d_h \times D_\ell}$	Row slice of $W_{\ell,b}^{O}$ for head h (rows $[(h-1)d_h,\ hd_h)$)
$\mathbf{b}_{\ell,b}^{O} \in \mathbb{R}^{D_\ell}$	Output projection bias at block (ℓ,b)
$\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M\times M}$	Relative position bias for head h at block (ℓ,b); shared across windows
$P:\mathbb{R}^{768}\to\mathbb{R}^{1024}$	CLAP audio projection head (two-layer MLP with residual)
Residual stream and head outputs	
$\mathbf{Z}^{(\ell,b)} \in \mathbb{R}^{S_\ell^2 \times D_\ell}$	Residual stream after block b of stage ℓ; see Eq. (5)
$\mathbf{A}_{\ell,b} \in \mathbb{R}^{N_w^\ell \cdot M \times D_\ell}$	W-MSA output at block (ℓ,b); see Eq. (12)
$\mathbf{M}_{\ell,b} \in \mathbb{R}^{S_\ell^2 \times D_\ell}$	MLP output at block (ℓ,b)
$\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$	Raw head output, pre-$W_{\ell,b}^{O}$; see Eq. (6)
$\widehat{\mathbf{H}}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times D_\ell}$	Per-head projected contribution, post-$W_{\ell,b}^{O}$; see Eq. (14)
Spatial aggregation and dataset matrices	
$\mathbf{r}_{\ell,b,h} \in \mathbb{R}^{d_h}$	Mean-pooled raw head output per sample; see Eq. (15)
$\widehat{\mathbf{r}}_{\ell,b,h} \in \mathbb{R}^{D_\ell}$	Mean-pooled projected head output per sample; see Eq. (**??**)
$\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n\times d_h}$	Dataset matrix of $\mathbf{r}_{\ell,b,h}$ across n samples
$\widehat{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{n\times D_\ell}$	Dataset matrix of $\widehat{\mathbf{r}}_{\ell,b,h}$ across n samples
n	Number of audio samples in the dataset
Semantic specialization	
$\mathbf{v}_1 \in \mathbb{R}^{d_h}$	First principal component of $\mathbf{R}_{\ell,b,h}$
$s_i = \mathbf{r}_{\ell,b,h}^{(i)} \cdot \mathbf{v}_1$	Score of sample i on the first principal component
$\boldsymbol{\sigma}_{\ell,b,h} \in \mathbb{R}^{D_{\text{CLAP}}}$	Semantic direction: variance-weighted combination of CLAP audio embeddings
$\mathbf{t}_c \in \mathbb{R}^{d_{\text{proj}}}$	ℓ_2-normalised text embedding of class c; fixed during optimisation
Z	Coherence Z-score of OMP-selected semantic atoms; see Eq. (**??**)
Zero-shot inference	
$\hat{\mathbf{a}}(x) \in \mathbb{R}^{d_{\text{proj}}}$	ℓ_2-normalised audio embedding of clip x
$\hat{y}(x) = \arg\max_c \hat{\mathbf{a}}(x)^\top \mathbf{t}_c$	Predicted class label; see Eq. (2)
Symbols introduced by RD* $(\star)$	
$\mathcal{L} \subseteq \{0,1,2,3\}$ $(\star)$	Target stage indices for reweighting; default $\{2,3\}$
$\rho_{\text{var}} \in (0,1)$ $(\star)$	Variance threshold for per-head subspace estimation (Eq. (35)); default 0.95
$k_{\ell,b,h} \in \{1,\ldots,d_h\}$ $(\star)$	Number of task-relevant PCA components; determined by ρ_{var}
$\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$ $(\star)$	Learnable per-component weight vector (`nn.Parameter`); init. via Eq. (36)
$\mathbf{C}_{\ell,b,h} \in \mathbb{R}^{d_h\times d_h}$	Empirical covariance of $\widetilde{\mathbf{R}}_{\ell,b,h}$; see Eq. (32)
$\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h\times d_h}$	Eigenvector matrix of $\mathbf{C}_{\ell,b,h}$; columns ordered by $\lambda_1^{\text{ev}} \geq \cdots \geq \lambda_{d_h}^{\text{ev}}$
λ_j^{ev}	j-th eigenvalue of $\mathbf{C}_{\ell,b,h}$; fixed after fitting, not trainable
$\boldsymbol{\mu}_{\ell,b,h} \in \mathbb{R}^{d_h}$	Empirical mean of $\widetilde{\mathbf{R}}_{\ell,b,h}$; fixed buffer, not trainable
$\widetilde{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{N_{\text{fit}}\times d_h}$	Fitting matrix; $N_{\text{fit}} = N_{\text{clips}} \cdot N_w^\ell \cdot M$; see Eq. (30)
$\widetilde{\mathbf{H}}_{\ell,b,h}, \mathbf{Z}_{\ell,b,h}, \widehat{\mathbf{Z}}_{\ell,b,h}$	Centred head output; PCA projection; rescaled projection (all $\in \mathbb{R}^{N_w^\ell \times M \times d_h}$)
$\mathbf{H}_{\ell,b,h}^{*} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$	Reweighted head output replacing $\mathbf{H}_{\ell,b,h}$ in Eq. (12); see Eq. (42)
$\mathrm{RD}^*(\mathbf{H}_{\ell,b,h}, \boldsymbol{\lambda}_{\ell,b,h})$	Full spectral reweighting transform at inference; see Eq. (44)
$\mathcal{J}(\boldsymbol{\lambda})$	Cross-entropy zero-shot loss; minimised over validation set; Eq. (37)
$\eta = 5\times10^{-3}, \gamma = 5\times10^{-2}$	AdamW learning rate and weight decay
$p=5, T_{\max}=40$	Early-stopping patience (epochs); maximum optimisation epochs
$N_{\max} = 10{,}000, N_{\text{PCA}} = 250$	Max fitting clips; samples for PCA fitting per fold

Table 4. **Selection of the most specialised heads (coherence Z-score).** ...

Head ID	**Layer**	**Dominant Cat.**	**Coherence Z**	**PC1 Var.**	**Primary Desc.**
`L3_B0_H7`	3	*human_body*	1.57	0.13	baby crying
`L2_B4_H1`	2	*human_body*	1.46	0.11	person sneezing
`L2_B5_H15`	2	*nature*	1.23	0.14	water dripping
`L3_B0_H25`	3	*nature*	1.19	0.19	water dripping
`L3_B1_H9`	3	*urban*	0.79	0.16	chainsaw cutting
`L2_B5_H6`	2	*animals*	0.71	0.14	frog croaking
`L3_B1_H17`	3	*animals*	0.65	0.29	rooster crowing

Table 5. Zero-shot accuracy (%) under 5-fold cross-validation for CLAP Classic and three ResiDual* configurations differing in the set of target stages $\mathcal{L}$. Mean $\pm$ standard deviation across folds. Bold indicates the highest accuracy for each dataset. Δ columns report the gain over CLAP Classic.

			$\mathcal{L} = \{3\}$		$\mathcal{L} = \{2,3\}$ (default)		$\mathcal{L} = \{0,1,2,3\}$	
Dataset	**Classes**	**CLAP Classic**	**RD***	Δ	**RD***	Δ	**RD***	Δ
ESC-50	50	93.85 ± 1.06	95.45	+1.60	$\mathbf{96.20 \pm 0.43}$	+2.35	94.85	+1.00
IRMAS	11	60.65 ± 1.57	79.55	+18.90	$\mathbf{81.80 \pm 1.69}$	+21.15	79.70	+19.05
TinySOL	14	35.55 ± 0.64	73.34	+37.79	$\mathbf{90.31 \pm 1.76}$	+54.76	89.91	+54.36
VocalSound	6	76.90 ± 1.25	86.30	+9.40	$\mathbf{88.30 \pm 1.77}$	+11.40	87.10	+10.20

Table 6. Per-fold zero-shot accuracy (%) for CLAP Classic and ResiDual* ($\mathcal{L} = \{2,3\}$). Bold entries indicate the higher accuracy within each fold and dataset.

Dataset	**Model**	**Fold 1**	**Fold 2**	**Fold 3**	**Fold 4**	**Fold 5**
ESC-50	CLAP Classic	92.25	95.00	93.75	93.25	95.00
	ResiDual*	**95.50**	**96.75**	**96.50**	**96.00**	**96.25**
IRMAS	CLAP Classic	59.50	60.00	62.75	62.25	58.75
	ResiDual*	**84.50**	**80.50**	**83.00**	**81.00**	**80.00**
TinySOL	CLAP Classic	35.24	35.24	34.96	35.53	36.78
	ResiDual*	**87.68**	**92.26**	**92.26**	**89.97**	**89.37**
VocalSound	CLAP Classic	74.75	77.00	76.50	78.00	78.25
	ResiDual*	**88.00**	**90.75**	**85.50**	**87.75**	**89.50**